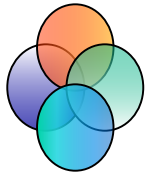
Maxwell's Equations (in diff. form)

Chapter 7 无穷算



传导电流

$$\oint_C \vec{H} \bullet d\vec{l} = \int_S (\vec{J} + \frac{\partial \vec{D}}{\partial t}) \bullet d\vec{S} \qquad \text{1st} \qquad \nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\oint_C \vec{E} \bullet d\vec{l} = -\int_S \frac{\partial \vec{B}}{\partial t} \bullet d\vec{S} \qquad \text{2nd} \qquad \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\oint_S \vec{D} \bullet d\vec{S} = Q \qquad \text{3rd} \qquad \nabla \bullet \vec{D} = \rho_{FC}$$

$$\oint_S \vec{B} \bullet d\vec{S} = 0 \qquad \text{4th} \qquad \nabla \bullet \vec{B} = 0$$

考研写形式默写

Manifesting relationship between fields & sources.

Material Equations

——auxiliary equations of Maxwell's Equs.

辅助方程

$$\begin{cases} \vec{D} = \varepsilon_0 \varepsilon_r \vec{E} = \varepsilon \vec{E} \\ \vec{B} = \mu_0 \mu_r \vec{H} = \mu \vec{H} \approx \mu_0 \vec{H} \\ \vec{J} = \sigma \vec{E} \end{cases}$$

选择题 三个之外的 不是

← 与极化相关的方程 Polarization

磁化

电阻

Maxwell's Eqs in Complex Form

For time harmonic field, 时谐.

$$\vec{E} = \vec{E}_0(x,y,z)\sin(\omega t + \varphi) \qquad \vec{B} = \vec{B}_0(x,y,z)\sin(\omega t + \phi)$$

In complex form, they can be expressed as,

$$\vec{E} = \vec{E}_e(x,y,z)e^{j\omega t} \qquad \vec{B} = \vec{B}_e(x,y,z)e^{j\omega t}$$

Therefore, we have $\dfrac{\partial}{\partial t} = j\omega, \quad \dfrac{\partial^2}{\partial t^2} = -\omega^2$

e.g. $\dfrac{\partial \vec{E}}{\partial t} = j\omega\vec{E}, \quad \dfrac{\partial^2 \vec{E}}{\partial t^2} = -\omega^2\vec{E}$

Maxwell's Eqs in Complex Form

For time harmonic field, $\frac{\partial}{\partial t} = j\omega, \quad \frac{\partial^2}{\partial t^2} = -\omega^2$

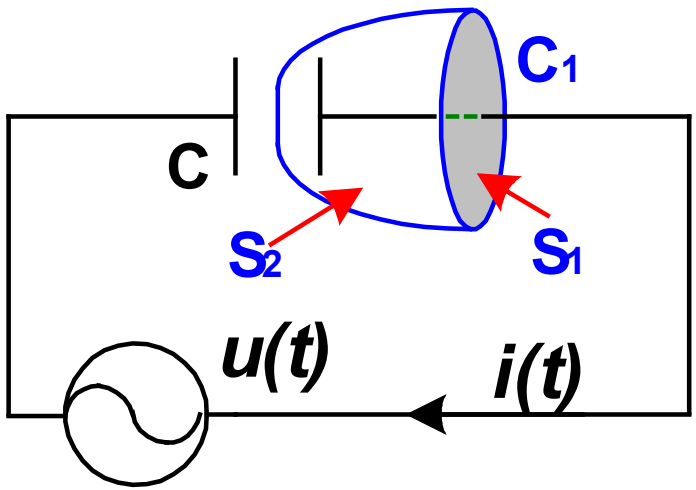
$$\begin{cases} \nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \\ \nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \\ \nabla \bullet \vec{B} = 0 \\ \nabla \bullet \vec{D} = \rho_{FC} \end{cases} \Rightarrow \begin{cases} \nabla \times \vec{\tilde{H}} = \vec{\tilde{J}} + j\omega \vec{\tilde{D}} \\ \nabla \times \vec{\tilde{E}} = -j\omega \vec{\tilde{B}} \\ \nabla \bullet \vec{\tilde{B}} = 0 \\ \nabla \bullet \vec{\tilde{D}} = \rho_{FC} \end{cases}$$

实际计算

- Purpose
  - For simplicity
  - For convenience of analysis in frequency domain

Density of Displacement Current

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t}$$



$$\oint_C \vec{H} \bullet d\vec{l} = \int_S \vec{J} \bullet d\vec{S}$$

$$\oint_C \vec{H} \bullet d\vec{l} = \int_S \left(\vec{J}_T + \vec{J}_d\right) \bullet d\vec{S}$$

$\vec{J}_T$ is density of conduction current.

The question is now answered satisfactorily

This is in fact Maxwell's 1st Equ..

表;辅;位移I
重点

jwD
jwεE

Boundary Conditions for E-Field

In normal direction

$$\oint_S \vec{D} \bullet d\vec{S} = Q$$

$$\vec{a}_n \bullet \left(\vec{D}_1 - \vec{D}_2\right) = \rho_{S_{FC}} \qquad \nabla \bullet \vec{D} = \rho_{FC}$$

$$D_{1n} - D_{2n} = \rho_{S_{FC}}$$

For boundary of ideal dielectrics $D_{1n} = D_{2n}$

For boundary of ideal conductor $D_{1n} = \rho_{S_{FC}}$

$$\oint_C \vec{E} \bullet d\vec{l} = -\int_S \frac{\partial \vec{B}}{\partial t} \bullet d\vec{S}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

In tangential direction

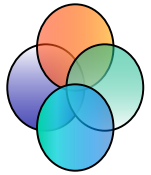
$$\vec{a}_n \times \left(\vec{E}_1 - \vec{E}_2\right) = 0$$

$$E_{1t} - E_{2t} = 0$$

For boundary of ideal dielectrics $E_{1t} = E_{2t}$

For boundary of ideal conductor $E_{1t} = 0$

Boundary Conditions for M-Field



In normal direction

$$\oint_S \vec{B} \bullet d\vec{S} = 0$$

$$\vec{a}_n \bullet \left(\vec{B}_1 - \vec{B}_2\right) = 0 \qquad \nabla \bullet \vec{B} = 0$$

$$B_{1n} - B_{2n} = 0$$

For boundary of ideal dielectrics $B_{1n} - B_{2n} = 0$

For boundary of ideal conductor $B_{1n} = B_{2n} = 0$

In tangential direction

$$\oint_C \vec{H} \bullet d\vec{l} = \int_S (\vec{J} + \frac{\partial \vec{D}}{\partial t}) \bullet d\vec{S}$$

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\vec{a}_n \times \left(\vec{H}_1 - \vec{H}_2\right) = \vec{J}_{ST}$$

$$H_{1t} - H_{2t} = J_{ST}$$

For boundary of ideal dielectrics $H_{1t} = H_{2t}$

For boundary of ideal conductor $H_{1t} = J_{ST}$

选考 6节

$$-\oint_S \left(\vec{E}\times\vec{H}\right)\bullet d\vec{S} = \int_V \frac{\partial}{\partial t}\left(\frac{1}{2}\varepsilon E^2 + \frac{1}{2}\mu H^2\right)dV + \int_V \sigma\cdot E^2 dV$$

闭合曲面 体积分

——Poynting's Law

$$-\oint_S \left(\vec{E}\times\vec{H}\right)\bullet d\vec{S} = \frac{\partial}{\partial t}\int_V \left(\frac{1}{2}\varepsilon E^2 + \frac{1}{2}\mu H^2\right)dV + \int_V (\sigma\cdot E^2)dV$$

左边矢量右边标量

E-engergy Density

M-energy Density

Variation Rate of Heat energy density

$$\oint_S \left(\vec{E}\times\vec{H}\right)\bullet d\vec{S} = -\frac{d}{dt}\left(W_e + W_m\right) - P_\Omega$$

E-power M-power Heat-power

Physical Meaning of Poynting's Law

$$\oint_S \left(\vec{E}\times\vec{H}\right)\bullet d\vec{S} = -\frac{d}{dt}\left(W_e + W_m\right) - P_\Omega$$

考物理意义

Power of energy flow out of closed surface

Energy Dissipating rate in closed surface

Define $\vec{S}_P = \vec{E}\times\vec{H}$ as Energy Flow Density.

Poynting's vector, *in honor of John H. Poynting*

注意量纲 Unit: W/m^2

$$\oint_S \vec{S}_P \bullet d\vec{S} = -\frac{d}{dt}\left(W_e + W_m\right) + \left(-P_\Omega\right)$$

Refer to power flow per cross area in EM-fields

(1) quantity: power density

(2) direction: normal to plane containing $\vec{E}$ & $\vec{H}$

Summary: Poynting's Vector

$$\vec{S}_P = \vec{E} \times \vec{H}$$ (unit: W/m^2)

Power flow vector, power flow density, EM-energy across per unit transect area during unit time.

$$\oint_S \vec{S}_P \bullet d\vec{S} = -\frac{d}{dt}\left(W_e + W_m\right) + \left(-P_\Omega\right)$$

Left side: power out of the closed surface at a certain moment.

Right side:

① dissipating rate of stored E-energy within that region

② dissipating rate of stored M-energy within that region

③ power of heat energy (ohmic power loss).

物理意义

A Comparison

考虑时P..，平均能流密度

$$-\oint_S \left(\vec{E}\times\vec{H}\right)\bullet d\vec{S} = \frac{\partial}{\partial t}\int_V \left(\frac{1}{2}\varepsilon E^2 + \frac{1}{2}\mu H^2\right)dV + \int_V (\sigma\cdot E^2)dV$$

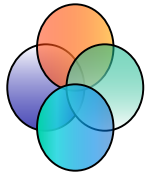
$$-\frac{1}{2}\oint_S \left(\vec{E}\times\vec{H}^*\right)\bullet d\vec{S} = jw\int_V \frac{1}{2}\left(\varepsilon E_0^2 - \mu H_0^2\right)dV + \int_V \frac{1}{2}(\sigma\cdot E^2)dV$$

Stored Energy　　Consumed Energy

- Poynting's Vector in Complex Form $\vec{S} = \frac{1}{2}(\vec{E}\times\vec{H}^*)$
- Average Poynting's Vector $\vec{S}_{av} = \frac{1}{2}\mathrm{Re}(\vec{E}\times\vec{H}^*)$

平均能流密度

§ 7.8 Scalar Potential & Vector Potential



- For static fields
  - Scalar potential $\vec{E} = -\nabla\psi$ Vector potential $\vec{B} = \nabla \times \vec{A}$
- For time-varying fields
  - Vector potential remains unchanged. However, curl of E is no longer 0 and scalar potential has to be modified.
  - Thus to define scalar potential we must find out a vector whose curl is 0.

$$\left.\begin{aligned} \nabla \times \vec{E} &= -\frac{\partial \vec{B}}{\partial t} \\ \vec{B} &= \nabla \times \vec{A} \end{aligned}\right\} \Rightarrow \quad \nabla \times \vec{E} = -\frac{\partial \nabla \times \vec{A}}{\partial t} = -\nabla \times \frac{\partial \vec{A}}{\partial t}$$

$$\Rightarrow \quad \nabla \times \left(\vec{E} + \frac{\partial \vec{A}}{\partial t}\right) = 0$$

$$-\nabla\psi = \vec{E} + \frac{\partial \vec{A}}{\partial t}$$

Vector in brackets can be expressed by the gradient of a scalar

不考，不常考

- Partial diff. Equ for vector potential

$$\nabla^2 \vec{A} - \mu\varepsilon \frac{\partial^2 \vec{A}}{\partial^2 t} = -\mu \vec{J}$$

- Partial diff. Equ for scalar potential

$$\nabla^2 \psi - \mu\varepsilon \frac{\partial^2 \psi}{\partial^2 t} = -\frac{\rho}{\varepsilon}$$

- Lorenz Condition & Lorenz Gauge
  - Significance: determine div of a vector potential

$$\nabla \bullet \vec{A} = -j\omega\mu\varepsilon\psi$$

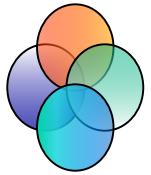
$$\psi = \frac{-\nabla \bullet \vec{A}}{j\omega\mu\varepsilon}$$

Coulomb's Gauge $\nabla \bullet \vec{A} = 0$ is just a special case.

$$\vec{E} = \frac{-\nabla(\nabla \bullet \vec{A})}{j\omega\mu\varepsilon} - j\omega\vec{A}$$

§ 8.1.2 Plane Wave

30'-7
30'-8
30'-9
10'-10



- Directions of mutually induced E- & M-fields are normal to each other.
- Vectors $\vec{E}$ & $\vec{H}$ constitute a surface, named as Constant Phase Surface等相位面.
  - Phase of E-field at any point on this surface is identical. The same is true to M-field.
  - direction of wave at a point is normal to constant phase surface.
- If constant phase surface is a plane, we call it a Plane Wave, or TEM wave (transverse electromagnetic wave 横电磁波).

横截

- In isotropic homogeneous media, the const phase surface is always a plane, and corresponding plane wave is named as the fundamental plane wave, or homogeneous plane wave (HPW)均匀平面波.
  - Spherical wave far away from its source can be .

General Characteristics & Expressions of Homogeneous Plane Wave (HPW)

(i) For homogeneous plane waves,

(a) Their polarizations are spacially uniform, it means their polarizations do not change and at the same time their magnitudes keep constant in the space, e.g. $E, H \rightarrow E_x, H_y$ or E_y, H_x

(b) They extend to infinity

(c) They exist only in homogeneous media

(ii) Therefore, they can be expressed as complex form

$$\vec{E}(\vec{r}) = \vec{E}_0 \exp[-j\vec{k} \cdot \vec{r}]$$

$$\vec{H}(\vec{r}) = \vec{H}_0 \exp[-j\vec{k} \cdot \vec{r}] \qquad (1)$$

where E_0 & H_0 are constant vectors,

加进去写

r is vector distance from the origin to a certain point,

k is constant vector known as wave vector and the time function e^{jwt} is omitted.

Vector Helmholtz Equations

For time harmonic field, $\frac{\partial}{\partial t} = j\omega, \quad \frac{\partial^2}{\partial t^2} = -\omega^2$

$$\begin{cases} \nabla^2 \vec{E} - \mu\varepsilon \dfrac{\partial^2 \vec{E}}{\partial t^2} = 0 \\ \nabla^2 \vec{H} - \mu\varepsilon \dfrac{\partial^2 \vec{H}}{\partial t^2} = 0 \end{cases} \Longrightarrow \begin{cases} \nabla^2 \tilde{\vec{E}} + \omega^2 \mu\varepsilon \tilde{\vec{E}} = 0 \\ \nabla^2 \tilde{\vec{H}} + \omega^2 \mu\varepsilon \tilde{\vec{H}} = 0 \end{cases}$$

Wave Equations　　Helmholtz Equations

Let $k^2 = \omega^2 \mu\varepsilon$

$$\begin{cases} \nabla^2 \tilde{\vec{E}} + k^2 \tilde{\vec{E}} = 0 \\ \nabla^2 \tilde{\vec{H}} + k^2 \tilde{\vec{H}} = 0 \end{cases}$$

HPW in Boundless Media

- Only incidence and no reflection

$$\left.\begin{aligned} E_x &= E_0 \mathrm{e}^{\mathrm{j}(\omega t - kz)} \\ H_y &= \left[E_0 / \sqrt{\mu/\varepsilon}\right] \mathrm{e}^{\mathrm{j}(\omega t - kz)} \end{aligned}\right\}$$

- Transient function

$$\left.\begin{aligned} E_x &= E_0 \sin(\omega t - kz) \\ H_y &= \left[E_0 / \sqrt{\mu/\varepsilon}\right] \sin(\omega t - kz) \end{aligned}\right\}$$

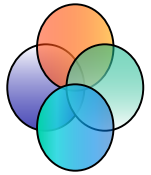
- Typical problems:
  - Given either E_* or H_*, please find the other, and also k & S.

Key to solution:

$$\nabla \times \vec{E} = -\mu \partial \vec{H} / \partial t \qquad \nabla^2 \tilde{\vec{H}} + k^2 \tilde{\vec{H}} = 0$$

$$\nabla \times \vec{H} = \varepsilon \partial \vec{E} / \partial t \qquad \nabla^2 \tilde{\vec{E}} + k^2 \tilde{\vec{E}} = 0$$

Example: Given E, Find H & S 必考



- For HPW, E is given as below. Please find H & S.

$$\vec{E} = \vec{e}_y E_y + \vec{e}_z E_z = \vec{e}_y E_0 \cos(\omega t - kx) + \vec{e}_z E_0 \sin(\omega t - kx)$$

Direction of EM-wave? $k_x^2 + k_y^2 + k_z^2 = k_x^2 + 0 + 0 = k^2 = \omega^2 \mu_0 \varepsilon_0$

Direction of M-field? Since it is TEM wave, H_x=0

$$\nabla \times \vec{E} = -\mu_0 \, \partial \vec{H} / \partial t$$

$$\nabla \times \vec{a} = \vec{e}_x \left(\frac{\partial a_z}{\partial y} - \frac{\partial a_y}{\partial z} \right) + \vec{e}_y \left(\frac{\partial a_x}{\partial z} - \frac{\partial a_z}{\partial x} \right) + \vec{e}_z \left(\frac{\partial a_y}{\partial x} - \frac{\partial a_x}{\partial y} \right)$$

$$-\mu_0 \frac{\partial H_x}{\partial t} = \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = 0$$

$$-\mu_0 \frac{\partial H_z}{\partial t} = \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}$$

$$-\mu_0 \frac{\partial H_y}{\partial t} = \frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x}$$

$$\vec{\boldsymbol{E}} = \vec{\boldsymbol{e}}_y E_y + \vec{\boldsymbol{e}}_z E_z = \vec{\boldsymbol{e}}_y E_0 \cos(\omega t - kx) + \vec{\boldsymbol{e}}_z E_0 \sin(\omega t - kx)$$

$t=0$ $x=0$ $\cos\cdots=1$ $\sin\cdots=0$ 逆时针 右旋圆极化

$$-\mu_0 \frac{\partial H_x}{\partial t} = \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} = 0 \qquad H_x = 0$$

$$-\mu_0 \frac{\partial H_y}{\partial t} = \frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} \qquad H_y = -E_0 \sqrt{\frac{\varepsilon_0}{\mu_0}} \sin(\omega t - kx) = -E_z / \eta_0$$

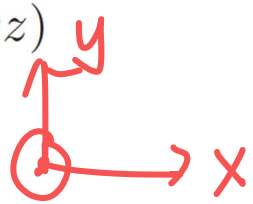
$$-\mu_0 \frac{\partial H_z}{\partial t} = \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \qquad H_z = E_0 \sqrt{\frac{\varepsilon_0}{\mu_0}} \cos(\omega t - kx) = E_y / \eta_0$$

$$\vec{\boldsymbol{H}} = \vec{\boldsymbol{e}}_y H_y + \vec{\boldsymbol{e}}_z H_z = -\vec{\boldsymbol{e}}_y \frac{E_0}{\eta_0} \sin(\omega t - kx) + \vec{\boldsymbol{e}}_z \frac{E_0}{\eta_0} \cos(\omega t - kx)$$

$$\vec{\boldsymbol{S}}(t) = \vec{\boldsymbol{E}} \times \vec{\boldsymbol{H}} = (\vec{\boldsymbol{e}}_y E_y + \vec{\boldsymbol{e}}_z E_z) \times (\vec{\boldsymbol{e}}_y H_y + \vec{\boldsymbol{e}}_z H_z) = \vec{e}_x E_0 H_0$$

7.2 Let the electric field intensity $\boldsymbol{E}$ of the electromagnetic wave in free space be

$$\boldsymbol{E} = \boldsymbol{e}_x E_0 \sin(\omega t - kz) + \boldsymbol{e}_y E_0 \sin(\omega t - kz)$$



Solve the following problems:

(1) the instantaneous magnetic field intensity $\boldsymbol{H}$.

(2) the poynting vector of the wave.

7.2 (1) $$\boldsymbol{H} = -\boldsymbol{e}_x \frac{E_0}{\eta_0} \sin(\omega t - kz) + \boldsymbol{e}_y \frac{E_0}{\eta_0} \sin(\omega t - kz)$$

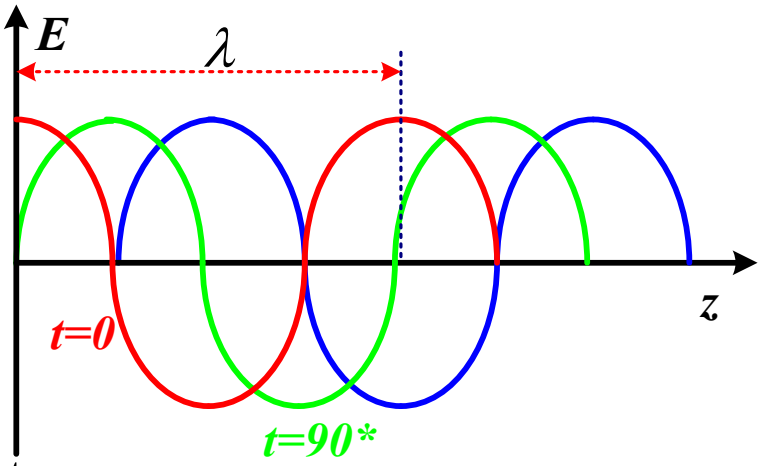
(2) Instantaneous value: $$\boldsymbol{S}(t) = \boldsymbol{e}_z 2 \frac{E_0^2}{\eta_0} \sin^2(\omega t - kz)$$

Mean value: $$\boldsymbol{S}_{\text{av}} = \boldsymbol{e}_z \frac{E_0^2}{\eta_0}$$

Key Parameters of HPW

$$E_x^+(z,t) = E_m^+ \cdot \cos(\omega t - kz)$$

Period $T = \dfrac{2\pi}{\omega}$ frequency $f = \dfrac{1}{T}$



Velocity of a single-f wave, or average v of a wave bundle

$$E_x^+(z,t) = E_m^+ \cos(\omega t - kz) = \text{const}$$

$$\omega t - kz = \text{const} \qquad \text{Take diff. as to } t\text{:} \qquad \omega - k \cdot \frac{dz}{dt} = 0$$

$$\therefore v = \frac{dz}{dt} = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\varepsilon}} = c\frac{1}{\sqrt{\mu_r\varepsilon_r}} = \frac{c}{n}$$

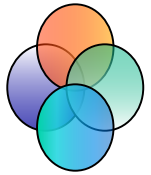
相移常数 物理意义

Wave Num. $k = \dfrac{\omega}{v} = \dfrac{2\pi}{\lambda}$ Unit: *rad/m*

Meaning: Phase-shift along per meter

1个 用入时名的..

Wave Impedance, Intrinsic Impedance, η



- Recall the solution to HPW

$$\left.\begin{aligned} E_x &= E_0 \sin(\omega t - kz) \\ H_y &= \left[E_0 / \sqrt{\mu/\varepsilon}\right] \sin(\omega t - kz) \end{aligned}\right\} \qquad \left.\begin{aligned} E_x &= E_0 \mathrm{e}^{\mathrm{j}(\omega t - kz)} \\ H_y &= \left[E_0 / \sqrt{\mu/\varepsilon}\right] \mathrm{e}^{\mathrm{j}(\omega t - kz)} \end{aligned}\right\}$$

$$\frac{E_x}{H_y} = \sqrt{\frac{\mu}{\varepsilon}} = \eta$$

This parameter is of unit of an impedance, and is called intrinsic impedance.

In free space, or air, we have

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi \approx 377(\Omega)$$

7.1 Let the instantaneous electric field intensity $\boldsymbol{E}$ of the electromagnetic wave in free space be

$$\boldsymbol{E} = \boldsymbol{e}_y 37.7 \cos(6\pi \times 10^8 t + 2\pi z)\,\text{V/m}$$

Solve the following problems:

(1) the direction of the wave propagation;

(2) the frequency, wavelength, phase constant and phase velocity;

(3) the instantaneous magnetic field intensity $\boldsymbol{H}$. Is it a uniform plane wave?

(1) The wave propagates in the direction of $-z$.

(2) $f = 300\text{MHz}$, $\lambda = 1\text{m}$, $k = 2\pi\text{rad/m}$, $v_\text{p} = 3 \times 10^8\text{m/s}$

(3) $\boldsymbol{H} = \boldsymbol{e}_x 0.1 \cos(6\pi \times 10^8 t + 2\pi z)\,\text{A/m}$

8.6 The electric field intensity of a uniform plane wave in free space is given by $\vec{\mathbf{E}} = 120\cos(2\pi \times 10^9 t - \beta y)\,\vec{\mathbf{a}}_z$ V/m. Determine (a) the phase constant, (b) the magnetic field intensity, (c) the wavelength, (d) the average power density in the medium, (e) the average energy density in the electric field, and (f) the average energy density in the magnetic field.

8.7 The magnetic field intensity of a plane wave in free space is given as $\vec{\mathbf{H}} = 0.1\cos(200\pi \times 10^6 t + \beta z)\,\vec{\mathbf{a}}_x$ A/m. Determine (a) the phase constant, (b) the velocity of propagation, (c) the wavelength, (d) the electric field intensity, (e) the displacement current density, and (f) the average power flow per unit area.

$\vec{H} \cdot \vec{S}_{av} = \vec{E}$

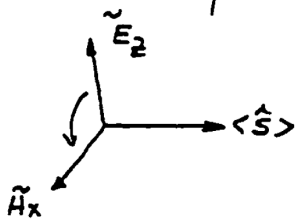
Exercise 8.6

$\omega = 2\pi \times 10^9$ rad/s

$\beta_0 = \frac{\omega}{c} = 20.94$ rad/m

$\eta_0 \cong 377\Omega$

$\vec{E} = 120\, e^{-j20.94y}\,\vec{a}_z$ V/m



For y-directed propagation,

$\vec{H} = 0.318\, e^{-j20.94y}\,\vec{a}_x$ A/m

$\langle \hat{S} \rangle = \frac{1}{2} Re\left[\tilde{E} \times \tilde{H}^*\right] = 19.1\,\vec{a}_y$ W/m^2

Exercise 8.7

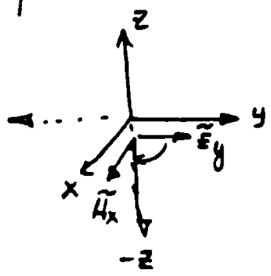
$\omega = 200\pi \times 10^6$ rad/s

$\beta_0 = \frac{\omega}{c} = \frac{2}{3}\pi$ rad/m, $\eta_0 = 377\Omega$

$\vec{u}_p = -3\times 10^8\,\vec{a}_z$ m/s

$\lambda_0 = \frac{2\pi}{\beta_0} = 3$ m

$\vec{H} = 0.1\, e^{j\beta_0 z}\,\vec{a}_x$ A/m



$\vec{E} = 37.7\, e^{j\beta_0 z}\,\vec{a}_y$ V/m

$\tilde{J}_d = j\omega\epsilon_0 \tilde{E} = j\,0.209\, e^{j\beta_0 z}\,\vec{a}_y$ A/m^2

$\langle \hat{S} \rangle = \frac{1}{2} Re\left[\tilde{E} \times \tilde{H}^*\right] = -1.885\,\vec{a}_z$ W/m^2

8.8 A 100-MHz uniform plane wave is traveling in the $+y$ direction in a lossless unbounded medium ($\epsilon_r = 4$ and $\mu_r = 1.0$). If the $\vec{\mathbf{E}}$ field has only an x component and its amplitude is 500 V/m when $t = 0$ and $y = 0$, determine (a) the phase velocity, (b) the phase constant, (c) the $\vec{\mathbf{H}}$ field, (d) the wavelength, and (e) the average power flow through a cross-sectional area of 16 cm^2.

$500\cos(\omega x - ky)$

Problem 8.8 $f = 100\ MHz$ $\omega = 2\pi f = 6.28\times10^8\ rad/s$ $\mu_r = 1$ $\epsilon_r = 4$

$\tilde{E}_x = 500\, e^{-j\beta y}\ V/m$ $u_p = \frac{c}{n} = 1.5\times10^8\ m/s$ $n = \sqrt{\mu_r \epsilon_r} = 2$

$\eta = \sqrt{\frac{\mu_0}{\epsilon_0}}\sqrt{\frac{\mu_r}{\epsilon_r}} = 60\pi\ \Omega$ $\beta = \frac{\omega}{u_p} = 4.19\ rad/m$

$\tilde{H}_z = -\frac{500}{60\pi} e^{-j\beta y}\ A/m$

$\langle \hat{S} \rangle = \frac{1}{2}\mathrm{Re}\left[\tilde{\vec{E}} \times \tilde{H}^*\right] = 663.15\ \vec{a}_y\ W/m$

$\langle P \rangle = \int_S \langle \hat{S} \rangle \cdot d\vec{s} = 1.06\ W$ = 平均能流密度 × 面积

EXAMPLE 8.1

WORKSHEET 7

If the electric field intensity as given by $\mathbf{E} = 377\cos(10^9 t - 5y)\vec{\mathbf{a}}_z$ V/m represents a uniform plane wave propagating in the y direction in a dielectric medium ($\mu = \mu_0$, $\epsilon = \epsilon_r \epsilon_0$), determine (a) the dielectric constant, (b) the velocity of propagation, (c) the intrinsic impedance, (d) the wavelength, (e) the magnetic field intensity, and (f) the average power density of the wave.

It is in +y direction

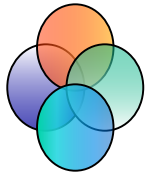
$$u_p = \frac{dy}{dt} = \frac{10^9}{5} = 2 \times 10^8 \text{ m/s}$$

$$v_p = \frac{1}{\sqrt{\mu_0 \varepsilon_0 \varepsilon_r}} = \frac{1}{\sqrt{\varepsilon_r}} \times \frac{1}{\sqrt{\varepsilon_0 \mu_0}} = \frac{1}{\sqrt{\varepsilon_r}} \times 3 \times 10^8 = 2 \times 10^8$$

$$so, \varepsilon_r = 2.25$$

EXAMPLE 8.2

The electric field intensity of a uniform plane wave in free space is given by $\vec{\mathbf{E}} = 94.25\cos(\omega t + 6z)\vec{\mathbf{a}}_x$ V/m. Determine (a) the velocity of propagation, (b) the wave frequency, (c) the wavelength, (d) the magnetic field intensity, and (e) the average power density in the medium.



Solution

a) The wave propagates in free space with the speed of light. Because the wave is traveling in the negative z direction, the phase velocity is

$$\vec{\mathbf{u}}_p = -3 \times 10^8\,\vec{\mathbf{a}}_z \text{ m/s}$$

b) $\beta_0 = 6$ rad/m, so the angular frequency of the wave is

$$\omega = \beta_0 u_p = 6 \times 3 \times 10^8 = 1.8 \times 10^9 \text{ rad/s}$$

c) The wavelength of the wave in free space is

$$\lambda_0 = \frac{2\pi}{\beta_0} = \frac{2\pi}{6} = 1.047 \text{ m}$$

d) The electric field intensity, in phasor form, is

$$\tilde{\vec{\mathbf{E}}} = 94.25e^{j6z}\,\vec{\mathbf{a}}_x \text{ V/m}$$

The corresponding $\vec{\mathbf{H}}$ field for the backward-traveling wave, from (8.27), is

$$\tilde{\vec{\mathbf{H}}} = -\frac{94.25}{377}\,e^{j6z}\,\vec{\mathbf{a}}_y = -0.25e^{j6z}\,\vec{\mathbf{a}}_y \text{ A/m}$$

or

$$\vec{\mathbf{H}}(z,t) = -0.25\cos(1.8 \times 10^9 t + 6z)\,\vec{\mathbf{a}}_y \text{ A/m}$$

e) The average power density in the medium is

$$\begin{aligned}\langle \hat{\mathbf{S}} \rangle &= \tfrac{1}{2}\text{Re}[\tilde{\vec{\mathbf{E}}} \times \tilde{\vec{\mathbf{H}}}^*] \\ &= -\tfrac{1}{2} \times 94.25 \times 0.25\,\vec{\mathbf{a}}_z \\ &= -11.78\,\vec{\mathbf{a}}_z \text{ W/m}^2\end{aligned}$$

electric field intensity in a dielectric ($\varepsilon_r = 4, \mu_r = 1$) as $\vec{E} = \vec{e}_z A\pi \cos(\omega t + \pi y)$

$$\beta = \pi \Rightarrow \lambda = \frac{2\pi}{\beta} = 2m$$

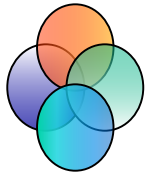
$$\beta = \pi = \omega\sqrt{\mu\varepsilon} \Rightarrow f = \frac{3\times 10^8}{2\sqrt{\varepsilon_r}} = 75\text{MHz}$$

$$\eta = \sqrt{\frac{\mu}{\varepsilon}} = \frac{120\pi}{\sqrt{\varepsilon_r}} = 60\pi\Omega$$

$$\vec{H} = \frac{-\vec{e}_y \times \vec{E}}{\eta} = -\vec{e}_x \frac{A}{60}\cos(\omega t + \pi y)$$

$$\vec{S}_{av} = -\vec{e}_y \frac{1}{2}\frac{E_0^2}{\eta} = -\vec{e}_y \frac{A^2\pi}{120}$$

Polarization?:



What is polarization?

Direction of E/M-field and its variation with time.

How to describe polarization?

Via the trace of terminal end of $\vec{E}$

Fundamental Polarization Type:

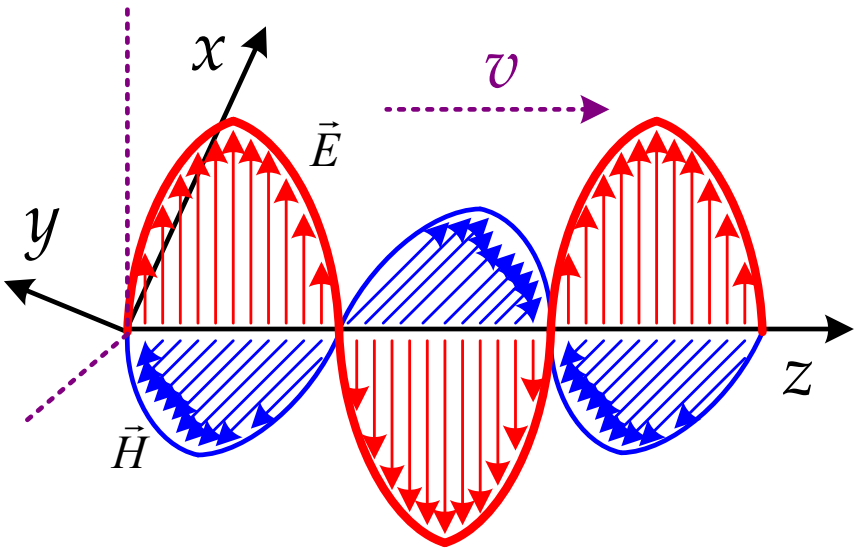
1. *Linear Polarization*
2. *Circular Polarization*
3. *Elliptical Polarization*

The other polarization types may be the combination of the above. And vice versa.

Check the combination of 2 single-polarized TEM waves

$$E_x = E_{xm}\cos(\omega t - kz - \varphi_x)$$

$$E_y = E_{ym}\cos(\omega t - kz - \varphi_y)$$



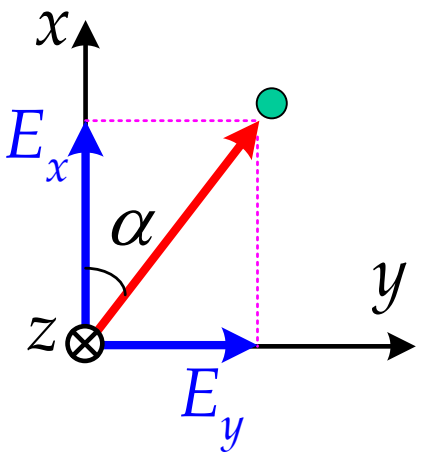
Sample the observation spot at z=0

$$E_x = E_{xm}\cdot\cos(\omega t - \varphi_x)$$

$$E_y = E_{ym}\cdot\cos(\omega t - \varphi_y)$$

Angle btwn overall E-field & x-axis

$$tg\alpha = \frac{E_y}{E_x} = \frac{E_{ym}\cos(\omega t - \varphi_y)}{E_{xm}\cos(\omega t - \varphi_x)}$$



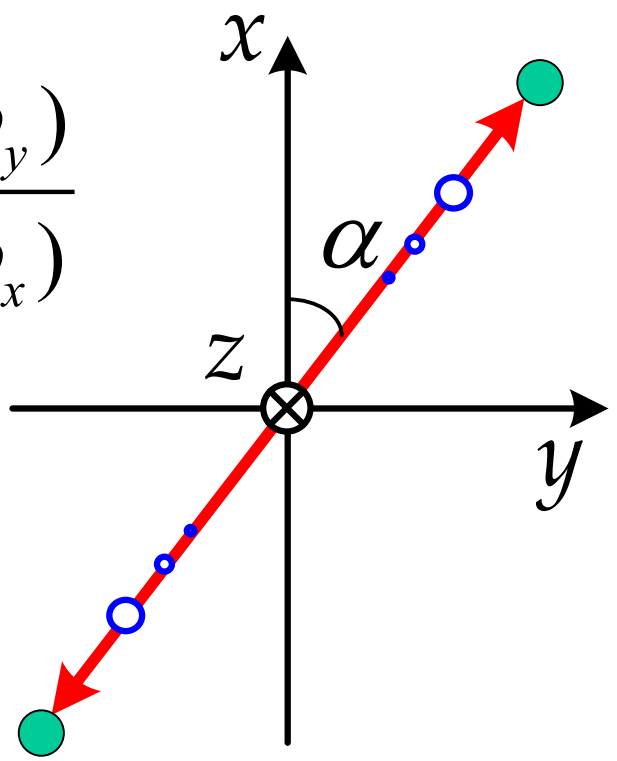
Linearly Polarized

Amplitude: $E_{xm}^2 + E_{ym}^2$

Phase: $tg\alpha = \dfrac{E_y}{E_x} = \dfrac{E_{ym}\cos(\omega t - \varphi_y)}{E_{xm}\cos(\omega t - \varphi_x)}$

If the trace of endpoint of E is a line,

$$tg\alpha = \frac{E_y}{E_x} = \text{Constant}$$

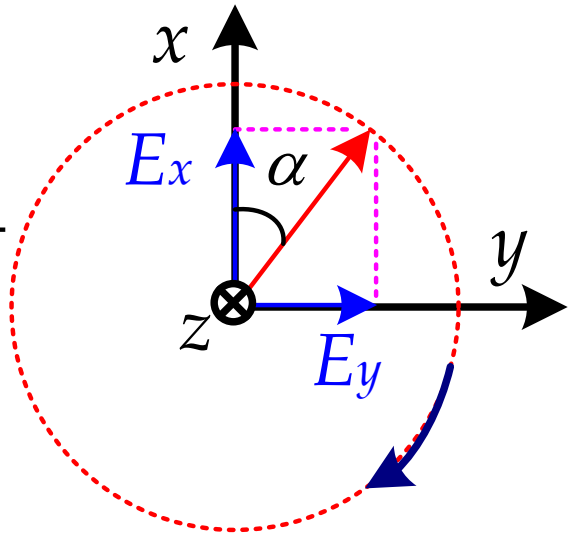


Condition: $\varphi_x = \varphi_y$ or $\varphi_x = \varphi_y + \pi$

Circularly Polarized

Amplitude: $E_{xm}^2 + E_{ym}^2$

Phase: $\tan\alpha = \dfrac{E_y}{E_x} = \dfrac{E_{ym}\cos(\omega t - \varphi_y)}{E_{xm}\cos(\omega t - \varphi_x)}$



Condition: $E_{xm} = E_{ym}$ and $\left|\varphi_x - \varphi_y\right| = \dfrac{\pi}{2}$

$E = \text{Constant}$ $\qquad tg\alpha = \dfrac{E_y}{E_x} = tg(\omega t)$ $\qquad \alpha = \omega t$

Sub-type: left-handed and right-handed circularly polarized

Judgment: put thumb to direction of wave, and check if the trace of endpoint of E-field is coincident with left hand or right hand.

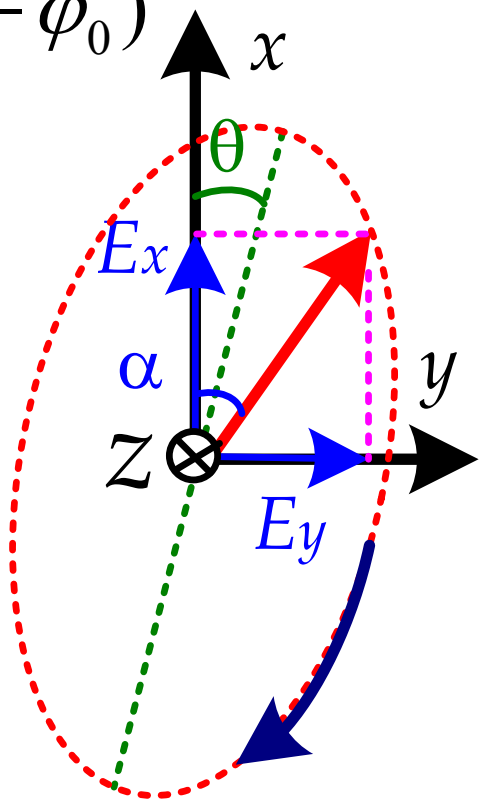
Elliptically Polarized

Condition: $E_{xm} \neq E_{ym}$ and $\varphi_y - \varphi_x = \varphi_0$

$$E_x = E_{xm} \cdot \cos(\omega t) \qquad E_y = E_{ym} \cdot \cos(\omega t - \varphi_0)$$

$$\frac{E_x^2}{E_{xm}^2} + \frac{E_y^2}{E_{ym}^2} - \frac{2 \cdot E_x \cdot E_x}{E_{xm} \cdot E_{ym}} \cdot \cos\varphi_0 = \sin^2 \varphi_0$$

An equation for ellipse



Angle btwn long axis & x-axis

$$tg(2\theta) = \frac{2E_{xm}E_{ym}}{E_{xm}^2 - E_{ym}^2} \cos\varphi_0$$

Example 1.

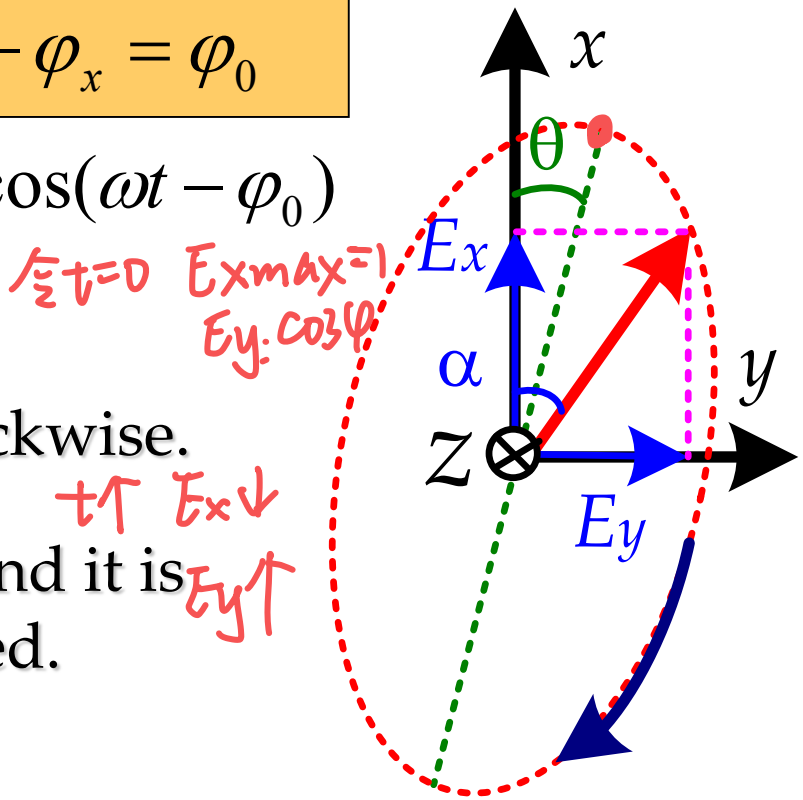
Condition: $E_{xm} \neq E_{ym}$ and $\varphi_y - \varphi_x = \varphi_0$

$$E_x = E_{xm} \cdot \cos(\omega t) \qquad E_y = E_{ym} \cdot \cos(\omega t - \varphi_0)$$

If φ_0>0, E_x ↓ and E_y ↑ with t ↑ .

Thus trace of endpoint of E is clockwise.

If TEM wave is in +z direction, we find it is right-handed elliptically polarized.



If φ_0<0, ……

If TEM wave is in –z direction, ……

Example 2. Problem 8.24

Please prove: a linearly polarized wave is the combination of a left-handed and a right-handed circularly polarized waves with the same amplitude.

Condition: $E_{xm} = E_{ym}$ and $\left|\varphi_x - \varphi_y\right| = \frac{\pi}{2}$

Solution: assume TEM wave is in +z direction and the amplitude of E is E_0. Thus 2 circularly polarized waves are

$$(\vec{a}_x E_0 - j\vec{a}_y E_0)e^{j(\omega t - kz)} \qquad \varphi_x - \varphi_y = \pi/2 > 0 \qquad \text{Right-handed}$$

$$(\vec{a}_x E_0 + j\vec{a}_y E_0)e^{j(\omega t - kz)} \qquad \varphi_x - \varphi_y = -\pi/2 < 0 \qquad \text{Left-handed}$$

$$\vec{E} = E_0\left[\left(\vec{a}_x - \mathrm{j}\vec{a}_y\right) + \left(\vec{a}_x + \mathrm{j}\vec{a}_y\right)\right]\cdot \mathrm{e}^{\mathrm{j}(\omega t - kz)} = \vec{a}_x(2E_0)\mathrm{e}^{\mathrm{j}(\omega t - kz)}$$

Their combination is linearly polarized.

Similar exercise --- Problem 8.25

7.15 试证明：

1) 一个椭圆极化波可以分解为旋转方向相反的两个圆极化波。

2) 求出这两个圆极化波的半径各为多少。

【解】1) 假设一个椭圆极化波为

$$\boldsymbol{E}=(\boldsymbol{e}_x A+\mathrm{j}\boldsymbol{e}_y B)$$

令 $A=A_1+A_2$，$B=A_1-A_2$，则

$$\boldsymbol{E}=(\boldsymbol{e}_x A+\mathrm{j}\boldsymbol{e}_y B)=(\boldsymbol{e}_x A_1+\mathrm{j}\boldsymbol{e}_y A_1)+(\boldsymbol{e}_x A_2-\mathrm{j}\boldsymbol{e}_y A_2)=\boldsymbol{E}_1+\boldsymbol{E}_2$$

显然，$\boldsymbol{E}_1$，$\boldsymbol{E}_2$ 都为圆极化波，但旋转方向相反。而且半径不等，分别为 A_1，A_2。

2) 由于 $A=A_1+A_2$，$B=A_1-A_2$，因此

$$A_1=\frac{A+B}{2},A_2=\frac{A-B}{2}$$

8.22 Find the polarization of the following waves:

a) $\tilde{\mathbf{E}} = 100e^{-j300x}\,\vec{\mathbf{a}}_y + 100e^{-j300x}\,\vec{\mathbf{a}}_z$ V/m

b) $\tilde{\mathbf{E}} = 16e^{j\pi/4}e^{-j100z}\,\vec{\mathbf{a}}_x - 9e^{-j\pi/4}e^{-j100z}\,\vec{\mathbf{a}}_y$ V/m

c) $\tilde{\mathbf{E}} = 3\cos(t - 0.5y)\,\vec{\mathbf{a}}_x - 4\sin(t - 0.5y)\,\vec{\mathbf{a}}_z$ V/m

$\theta_x - \theta_y = \frac{\pi}{2}$

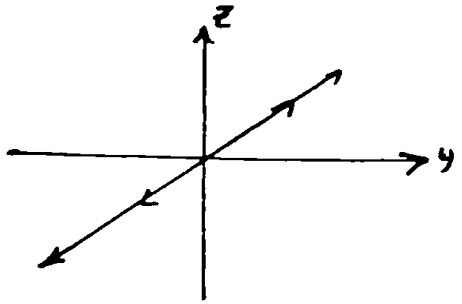
$9e^{j\frac{\pi}{4}}$

Problem 8.22 a) At $x=0$

$E_y = 100\cos\omega t$

$E_z = 100\cos\omega t$ $\Rightarrow E_y = E_z$

Linear polarization



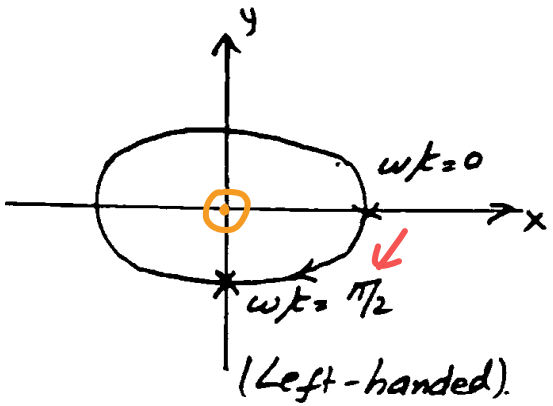
b. When $100z = \pi/4$

振幅不同

$E_x = 16\cos\omega t$

$E_y = -9\sin\omega t$ $\Rightarrow \left(\frac{E_x}{16}\right)^2 + \left(\frac{E_y}{9}\right)^2 = 1$

(Elliptical)



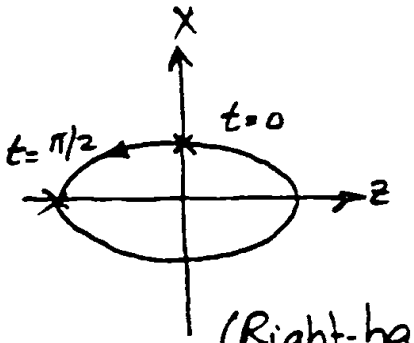
(Left-handed)

c) In a $y=0$ plane:

$E_x = 3\cos t$

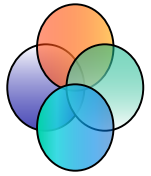
$E_z = -4\sin t$ $\Rightarrow \left(\frac{E_x}{3}\right)^2 + \left(\frac{E_z}{4}\right)^2 = 1$

Elliptical



(Right-handed)

8.23 A uniform plane wave with an $\vec{\mathbf{E}}$ of $12\cos(\omega t - \beta z)\,\vec{\mathbf{a}}_x - 5\sin(\omega t - \beta z)\,\vec{\mathbf{a}}_y$ V/m is propagating in a lossless medium ($\epsilon_r = 2.5$, $\mu_r = 1$) at 200 Mrad/s. Determine the corresponding $\vec{\mathbf{H}}$ field, the phase constant β, the wavelength λ, the intrinsic impedance η, the phase velocity $\vec{\mathbf{u}}_p$, and the polarization of the wave.



Problem 8.23

$\omega = 2\times10^8$ rad/s

$\epsilon_r = 2.5$

$\lambda = \frac{2\pi}{\beta} = 5.96$ m

$\beta = \frac{\omega}{c}\sqrt{\epsilon_r} = 1.054$ rad/m

$\eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{120\pi}{\sqrt{\epsilon_r}} = 238.43\,\Omega$

Use Superposition Theorem:

$\tilde{E}_x = 12\,e^{-j\beta z} \Rightarrow \tilde{H}_y = \frac{12}{\eta}e^{-j\beta z}$

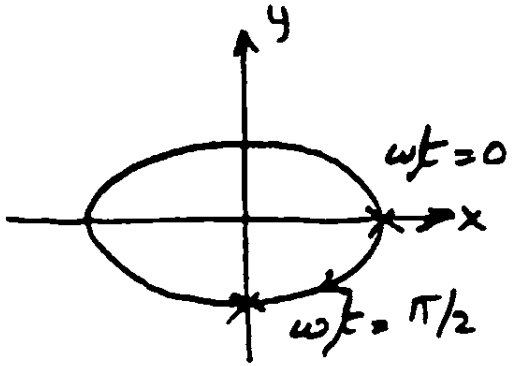
$\tilde{E}_y = j5\,e^{-j\beta z} \Rightarrow \tilde{H}_x = -j\frac{5}{\eta}e^{-j\beta z}$

$\tilde{\vec{H}} = \frac{1}{\eta}\left[12\,e^{-j\beta z}\vec{a}_y - j5\,e^{-j\beta z}\vec{a}_x\right]$

$\vec{H} = 0.05\cos(\omega t - \beta z)\,\vec{a}_y + 0.02\sin(\omega t - \beta z)\,\vec{a}_x$ A/m

at $z=0$ $E_x = 12\cos\omega t$, $E_y = -5\sin\omega t$

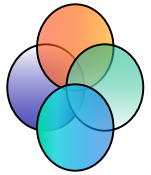
$\left(\frac{E_x}{12}\right)^2 + \left(\frac{E_y}{5}\right)^2 = 1$ Elliptical.



Left-handed

10 HPW in conducting media

E_x H_y



$$\nabla^2\vec{E}-\gamma^2\vec{E}=0 \Longrightarrow \frac{\partial^2 E_x}{\partial z^2}-\gamma^2 E_x=0$$

$$\nabla^2\vec{H}-\gamma^2\vec{H}=0 \Longrightarrow \frac{\partial^2 H_y}{\partial z^2}-\gamma^2 H_y=0$$

Combine both incident and reflected waves

$$E_x=f\left(t-\frac{z}{v}\right)+f\left(t+\frac{z}{v}\right)$$

+z direction -z direction

$$E_x=E_0 e^{j\omega t-\gamma z}+E_0' e^{j\omega t+\gamma z}$$

Similar to scalar solution to wave equ in perfect dielectrics

$$E_x=E_0 e^{j(\omega t-kz)}+E_0' e^{j(\omega t+kz)}$$

(2) Propagation Const. $\gamma = \sqrt{\mathrm{j}\omega\mu\sigma - \omega^2\mu\varepsilon}$ $\gamma=\alpha+\mathrm{j}\beta$

$$E_x = E_0 \mathrm{e}^{\mathrm{j}\omega t - \gamma z} \qquad E_x = E_0 \cdot \mathrm{e}^{-\alpha z} \cdot \mathrm{e}^{\mathrm{j}(\omega t - \beta z)}$$

α is called the attenuation const., refers to the loss of the amplitude per meter along propagating direction NP/m

β is the phase-shift const., means the phase change of HPW per meter along propagating direction

(3) Phase Velocity v

$$v_p(z) = \frac{\omega}{\beta} = 1 \Big/ \sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1+\frac{\sigma^2}{\omega^2\varepsilon^2}}+1\right)}$$

Different medium can be classified by the loss tangent $\tan\delta$.

Perfect conductor: $\tan\delta = \dfrac{\sigma}{\omega\varepsilon'} \to \infty;$

Good conductor: $\tan\delta = \dfrac{\sigma}{\omega\varepsilon'} \gg 1;$

A good dielectric is a dielectric material which satisfies $\dfrac{\partial}{\omega\varepsilon'} \ll 1$

§ 8.5.2 HPW in Good Conductors

$\sigma/\omega\varepsilon$ 与1判断 → $\dfrac{\sigma}{\omega\varepsilon} >> 1$

$$E_x = E_0 e^{j\omega t - \gamma z}$$

$$H_y = \left(\frac{\gamma}{j\omega\mu}\right) E_0 e^{j\omega t - \gamma z}$$

$\gamma = \alpha + j\beta$

$$E_x = E_0 e^{-\alpha z} \cdot e^{j(\omega t - \beta z)}$$

$$H_y = \frac{1}{\eta^e} E_0 e^{-\alpha z} \cdot e^{j(\omega t - \beta z)}$$

$$\gamma = \sqrt{j\omega\mu\sigma - \omega^2\mu\varepsilon} = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)} \approx \sqrt{j\omega\mu\sigma}$$

$$\gamma = \alpha + j\beta \approx \sqrt{j\omega\mu\sigma} = \sqrt{\pi f\mu\sigma} + j\sqrt{\pi f\mu\sigma}$$

$$\alpha \approx \sqrt{\pi f\mu\sigma}$$

$$\beta \approx \sqrt{\pi f\mu\sigma}$$

$$\eta^e = \sqrt{\frac{\mu}{\varepsilon^e}} = \sqrt{\frac{\mu}{\varepsilon(1 - j\frac{\sigma}{\omega\varepsilon})}} \approx \sqrt{j\frac{\omega\mu}{\sigma}} = \sqrt{\frac{\omega\mu}{\sigma}} e^{j45^\circ}$$

Phase Diff btwn E&M is 45°.

Velocity: $$v = \frac{\omega}{\beta} = \frac{\omega}{\sqrt{\pi f \mu \sigma}} = \sqrt{\frac{2\omega}{\mu\sigma}}$$

Velocity of EM-wave in good conductor is a function of w, and thus we call this phenomenon Dispersion (色散).

Notice that better conductor yields slower velocity (due to a larger σ).

Skin Effect & Skin Depth

EM fields or the induced current attenuates along z-axis according to the rule of $e^{-\alpha z}$.

Therefore, the time-varying fields or alternating current concentrates only in a thin layer next to the outer surface of the conductor. This phenomenon is called skin effect.

When the magnitude attenuates to $1/e$ of its original value, we call the depth as the skin depth or the penetration depth (趋肤深度 δ).

考

$$E_x = E_0 \mathrm{e}^{-\alpha z} \cdot \mathrm{e}^{\mathrm{j}(\omega t - \beta z)}$$

$$E_{xm} = E_0 \mathrm{e}^{-\alpha z}$$

$$E_0 \mathrm{e}^{-\alpha \delta} = E_0 / e$$

$$\delta = \frac{1}{\alpha} = \frac{1}{\omega \sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2} - 1\right)}}$$

skin depth

$$\delta = \frac{1}{\alpha} = \frac{1}{\omega\sqrt{\frac{\mu\varepsilon}{2}\left(\sqrt{1+\left(\frac{\sigma}{\omega\varepsilon}\right)^2}-1\right)}}$$

Good conductor

$$\approx \frac{1}{\sqrt{\pi f \mu \sigma}}$$

For good conductor

$$\alpha = \beta = \sqrt{\pi f \mu \sigma} \qquad \beta = 2\pi/\lambda \qquad \delta = 1/\alpha \qquad \therefore \delta = \frac{\lambda}{2\pi}$$

Skin depth is the depth that an EM-wave penetrates into the good conductor effectively.

Better conductor, $\sigma \uparrow, \delta \downarrow$. Also, $f \uparrow, \delta \downarrow$.

交变电磁场进入导体表面后很快就衰减殆尽,“势力范围”只在离表面很浅的导体中,顾名思义”趋肤深度”。

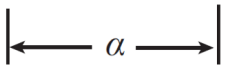
- 对于同一块导体，其交流（AC）电阻率$(1/\sigma\delta)$比直流（DC）电阻率$(1/\sigma)$大，这是趋肤效应所造成的。
  - 解释为：对高频电流，由于趋肤效应，与均匀分布在导体中的直流电流相比较，其有效的导电面积大大的减少，电阻增大。
- R_S是在假定导体的厚度为无穷大的条件下得到的，
  - 对于厚度有限的实际导体，上式精确度很高；
  - 对于圆柱形导体，把圆柱纵向视为长度，圆周视为宽度，径向视为厚度，上式依然适用.

例： 书P489 例9.19

7.16 Generally, in order to realize effective electromagnetic shielding, the thickness of shielding layer is set to be one wavelength. Find

(1) the thickness of Aluminum ($\sigma = 10^7$S/m, $\varepsilon_r = 1$, $\mu_r = 10^4$) shielding in intermediate frequency transformer of radio (The frequency is 465kHz).

(2) the thickness of iron ($\sigma = 10^7$S/m, $\varepsilon_r = 1$, $\mu_r = 10^4$) shielding in power transformer (The frequency is 50Hz).



since $\frac{\sigma}{\omega\varepsilon} >> 1$, it is a good conductor

$\beta \approx \sqrt{\pi f \mu \sigma}$, $d = 2\pi/\beta$.

$d = \lambda$

7.16 (1) $d = 0.76$mm

(2) $d = 1.414$mm

7.20 The conductivity of sea water is $\sigma = 4\text{S/m}$ and $\varepsilon_r = 8$. Find the attenuation constant, wavelength and wave impedance of the electromagnetic wave in the sea with frequencies of 10kHz, 1MHz, 10MHz and 1GHz.

$$\text{since } \frac{\sigma}{\omega\varepsilon} >> 1, \text{ it is a good conductor}$$

$$\alpha \approx \sqrt{\pi f \mu \sigma}, \beta \approx \sqrt{\pi f \mu \sigma}, \ \lambda = 2\pi / \beta.$$

7.20 When $f = 10\text{kHz}$, $\alpha = 0.126\pi$, $\lambda = 15.87\text{m}$, $\eta = 0.0316\pi(1+j)(\Omega)$

When $f = 1\text{MHz}$, $\alpha = 1.26\pi, \lambda = 1.587\text{m}$, $\eta = 0.316\pi(1+j)(\Omega)$

When $f = 10\text{MHz}$, $\alpha = 4\pi, \lambda = 0.5\text{m}$, $\eta = \pi(1+j)(\Omega)$

When $f = 1\text{GHz}$, $\alpha = 24.65\pi, \beta = 2\pi \times 32.4\text{rad/m}$, $\lambda = 0.03\text{m}$, $\eta = 42/\sqrt{1 - j \times 0.89}(\Omega)$

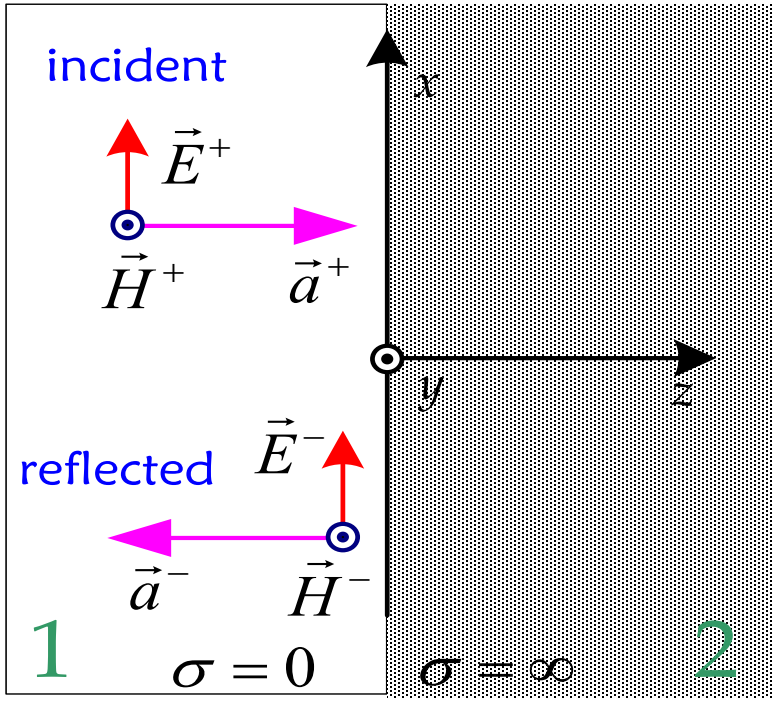
9.1 Normal incidence on surface of perfect conductor

Incident wave in region 1

$$\vec{E}^{+} = \vec{a}_x E_0^{+} e^{j(wt-kz)}$$

$$\vec{H}^{+} = \vec{a}_z \times \vec{E}^{+} / \eta_1$$

$$\vec{\boldsymbol{H}}^{+} = \vec{a}_y H_y^{+} = \vec{a}_y \frac{E_0^{+}}{\eta} \mathrm{e}^{\mathrm{j}(\omega t - kz)}$$



Reflected wave in region 1

$$\vec{\boldsymbol{E}}^{-} = \vec{a}_x E_0^{-} \mathrm{e}^{\mathrm{j}(\omega t + kz)}$$

$$\vec{H}^{-} = (-\vec{a}_z) \times \vec{E}^{-} / \eta$$

$$\vec{\boldsymbol{H}}^{-} = \vec{a}_y H_0^{-} \mathrm{e}^{\mathrm{j}(\omega t + kz)} = -\vec{a}_y (E_0^{-} / \eta) \mathrm{e}^{\mathrm{j}(\omega t + kz)}$$

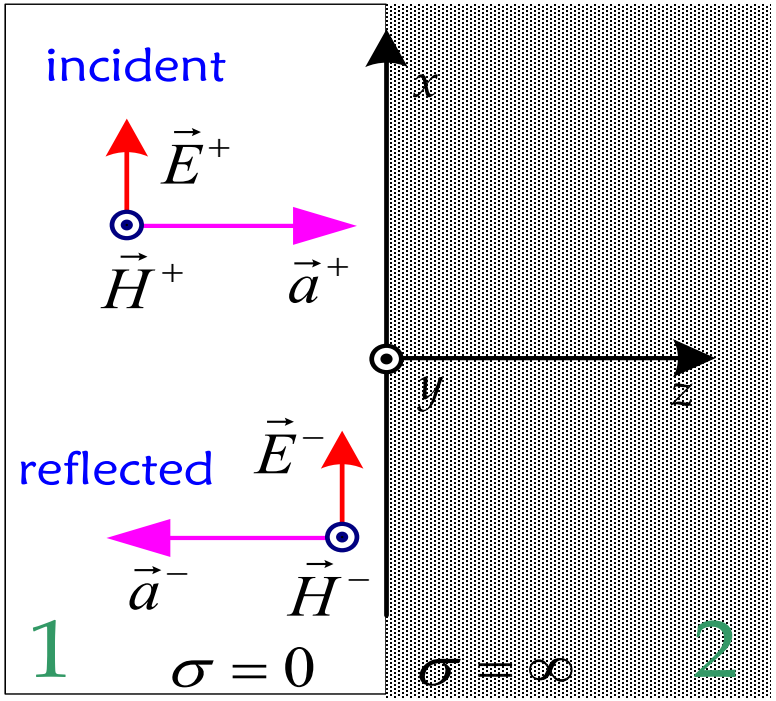
Total E-Field

incident $\vec{E}^{+} = \vec{a}_x E_0^{+} e^{j(\omega t - kz)}$

reflected $\vec{E}^{-} = \vec{a}_x E_0^{-} \mathrm{e}^{\mathrm{j}(\omega t + kz)}$

total $\vec{E}_1 = \vec{E}^{+} + \vec{E}^{-}$

$$= \vec{a}_x \left(E_0^{+} e^{-jkz} + E_0^{-} e^{+jkz} \right) e^{jwt}$$



Boundary conditions:

$$E_{1t}\big|_{z=0} = E_{2t}\big|_{z=0} = E_0^{+} + E_0^{-} = 0$$

$$\therefore E_0^{+} = -E_0^{-}$$

$$\vec{E}^{-} = -\vec{a}_x E_0^{+} \mathrm{e}^{\mathrm{j}(\omega t + kz)}$$

$$\vec{E}_1 = -\vec{a}_x \mathrm{j}2E_0^{+} \sin(kz) \cdot \mathrm{e}^{\mathrm{j}\omega t}$$

Total M-Field

incident $\vec{H}^+ = \vec{a}_y (E_0^+ / \eta) \mathrm{e}^{\mathrm{j}(\omega t - kz)}$

reflected $\vec{H}^- = -\vec{a}_y (E_0^- / \eta) \mathrm{e}^{\mathrm{j}(\omega t + kz)}$

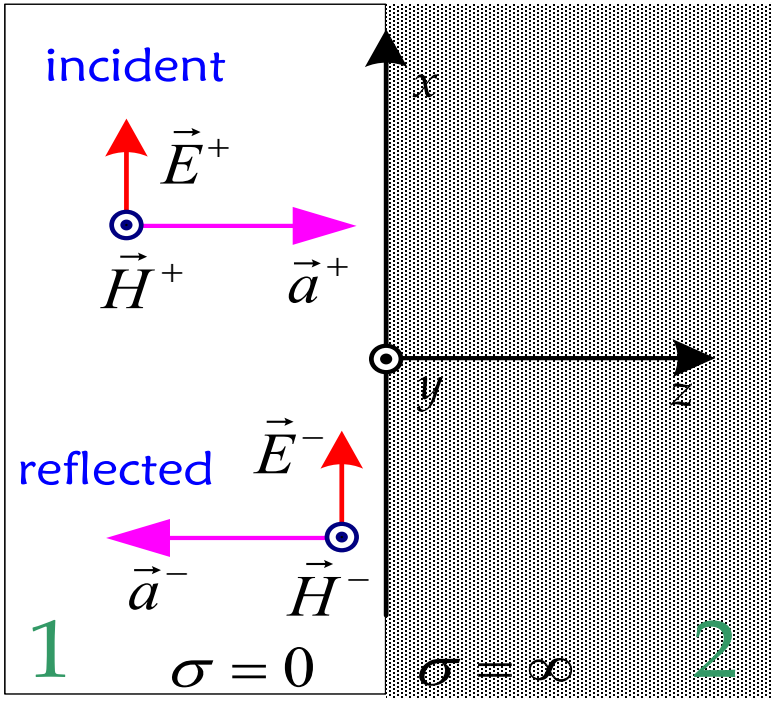
$= \vec{a}_y (E_0^+ / \eta) \mathrm{e}^{\mathrm{j}(\omega t + kz)}$

total

$$\vec{H}_1 = \vec{H}^+ + \vec{H}^-$$

$$= \vec{a}_y \left(\frac{E_0^+}{\eta} \mathrm{e}^{-\mathrm{j}kz} + \frac{E_0^+}{\eta} \mathrm{e}^{\mathrm{j}kz} \right) \mathrm{e}^{\mathrm{j}\omega t}$$

$$= \vec{a}_y 2 \frac{E_0^+}{\eta} \cos(kz) \cdot \mathrm{e}^{\mathrm{j}\omega t} = \vec{a}_y 2 H_0^+ \cos(kz) \mathrm{e}^{\mathrm{j}\omega t}$$



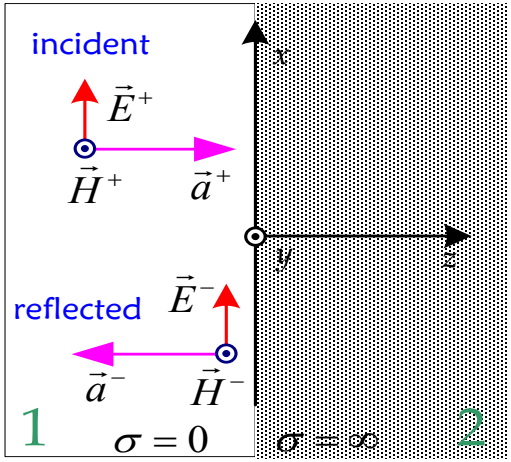
☆ 考

+ 考表面电流

At boundary

$$\vec{E}_1\big|_{z=0} = 2E_0^+ \sin\left(\beta_1 z\big|_{z=0}\right)\sin(\omega t) = 0$$

$$\vec{H}_1\big|_{z=0} = \frac{2E_0^+}{\eta_1}\cos\left(\beta_1 z\big|_{z=0}\right)\cos(\omega t) = \frac{2E_0^+}{\eta_1}\cos(\omega t)$$



Induced surface current

$$\vec{J}_{S_{FC}} = \vec{a}_n \times \vec{H}_1\big|_{z=0} = \left(-\vec{a}_z\right)\times\vec{a}_y\ldots = \vec{a}_x\ldots$$

$$\vec{J}_{S_{FC}} = \vec{a}_x \frac{2E_0^+}{\eta_1}\cos(\omega t)$$

Average Poyting's vector

$$\vec{P}_{av} = \vec{S}_{av} = \frac{1}{2}R_e\left(\vec{E}\times\vec{H}^*\right) = 0$$

$$\vec{E}\times\vec{H}^* = \left[\vec{a}_x\left(-j2E_0^+\sin(\beta_1 z)\right)\right]\times\left[\vec{a}_y\left(\frac{2E_0^+}{\eta_1}\cos(\beta_1 z)\right)\right]$$

Standing wave --- no energy transmitted

8.1 A uniform plane wave is incident from air normally upon the surface of perfect conductor at $z = 0$ (xOy plane), given the electric field of incident wave $E_x^+ = E_0^+ e^{j(\omega t - kz)}$, try to find:

(1) magnetic field of incident wave H_y^+;

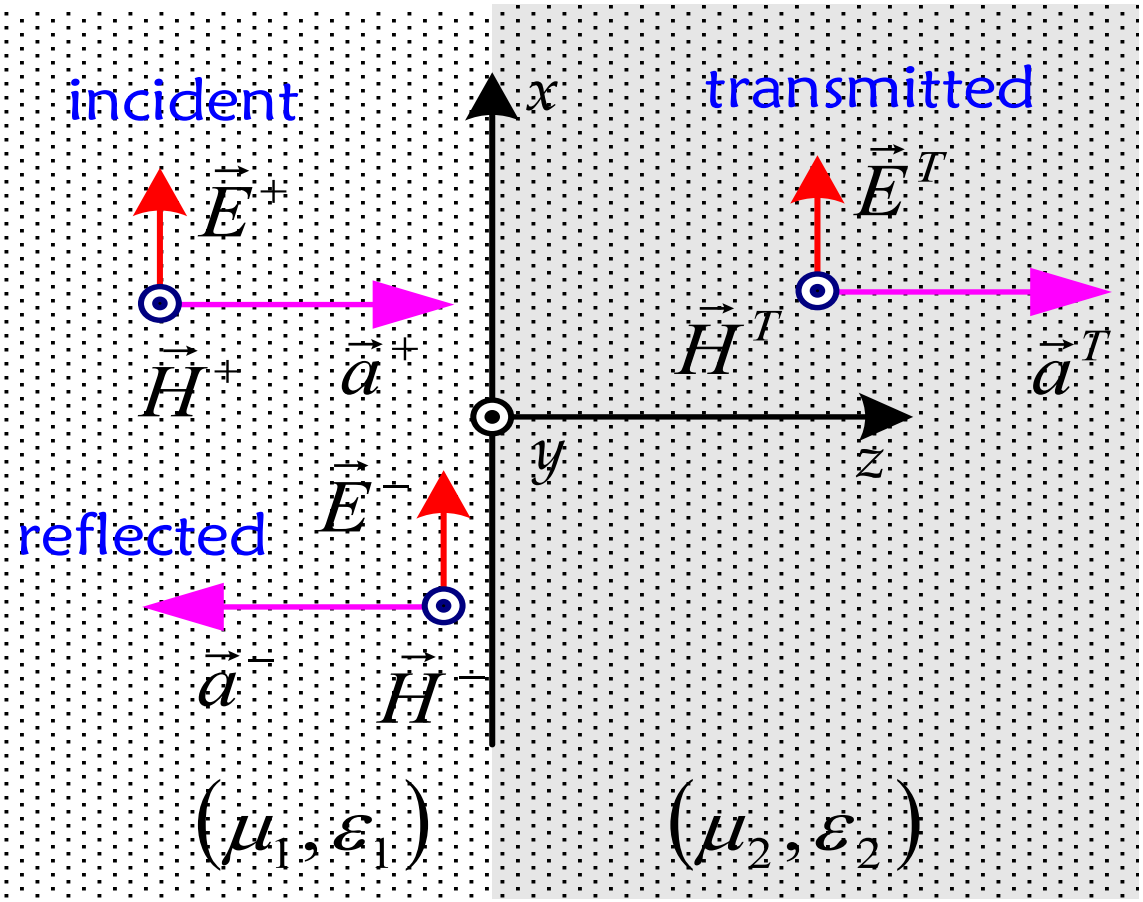
(2) magnetic field of reflected wave H_y^-;

(3) magnetic field of the total wave $H_y = H_y^+ + H_y^- = ?$

8.1 (1) $H_y^+ = \frac{k}{\omega\mu_0} E_x^+ = \left(\frac{1}{\eta_0}\right) E_x^+ = \frac{1}{\eta_0} E_0^+ e^{j(\omega t - kz)}$ (2) $H_y^- = \left(\frac{1}{\eta_0}\right) E_0^+ e^{j(\omega t + kz)}$

(3) $H = 2\left(\frac{1}{\eta_0}\right) E_0^+ \cos(kz) e^{j\omega t}$

9.2 Normal incidence on surface of perfect dielectric



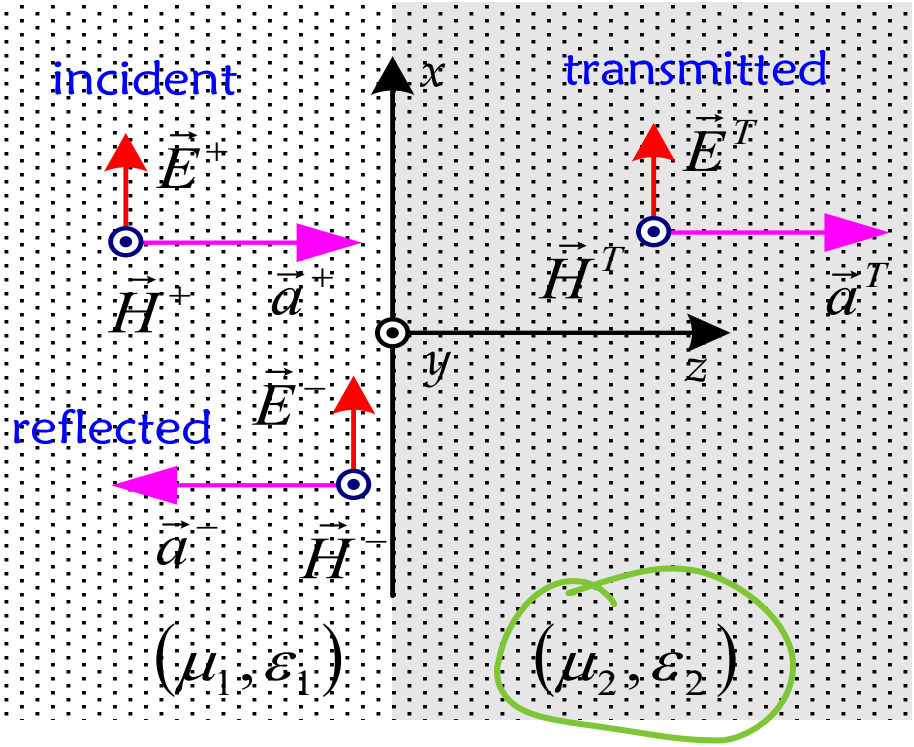
Incident Wave

assume

$$\vec{E}^{+} = \vec{a}_x E_x^{+} = \vec{a}_x E_0^{+} \mathrm{e}^{\mathrm{j}(\omega t - k_1 z)}$$

$$\vec{H}^{+} = \vec{a}_y H_y^{+} = \vec{a}_y H_0^{+} \mathrm{e}^{\mathrm{j}(\omega t - k_1 z)}$$

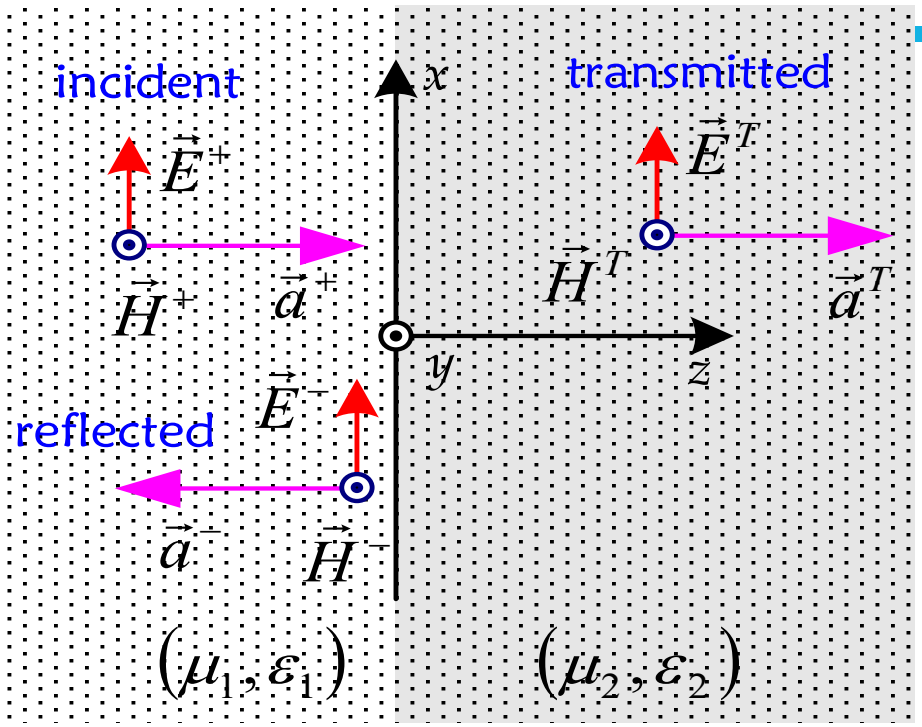
$$= \vec{a}_y \frac{E_0^{+}}{\eta_1} \mathrm{e}^{\mathrm{j}(\omega t - k_1 z)}$$



$k_1 = \omega\sqrt{\mu_0 \varepsilon_1}$ Propagation const. in region 1

$\eta_1 = \sqrt{\mu_0 / \varepsilon_1}$ Intrinsic impedance in region 1

Reflected Wave



$$\vec{E}^{-} = \vec{a}_x E_x^{-} = \vec{a}_x E_0^{-} \mathrm{e}^{\mathrm{j}(\omega t + k_1 z)}$$

By comparison

$$\vec{E}^{+} = \vec{a}_x E_x^{+} = \vec{a}_x E_0^{+} \mathrm{e}^{\mathrm{j}(\omega t - k_1 z)}$$

$$\vec{H}^{-} = \vec{a}_y H_0^{-} \mathrm{e}^{\mathrm{j}(\omega t + k_1 z)}$$

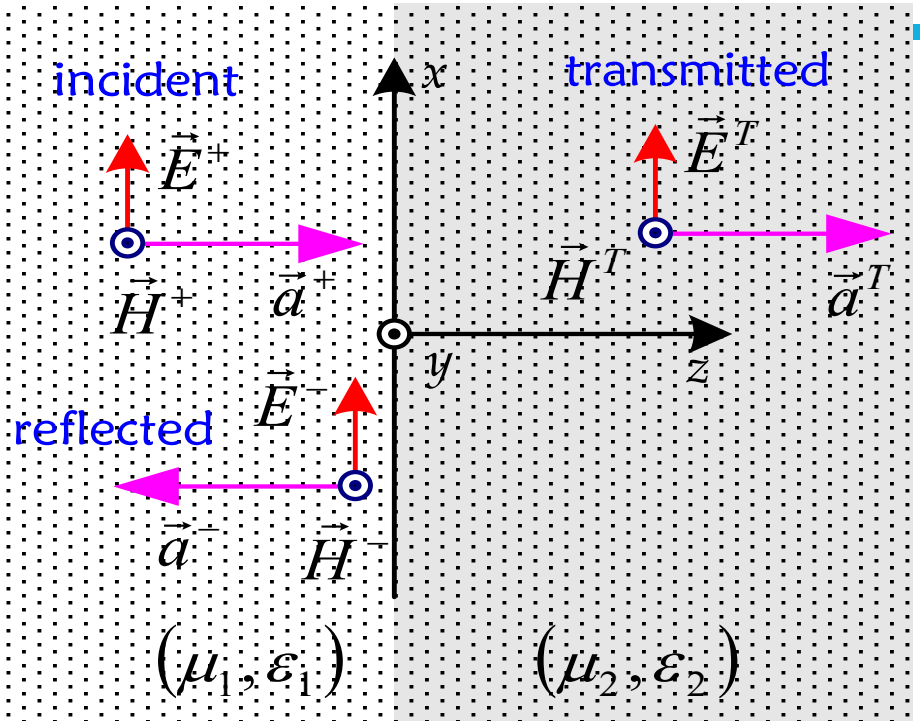
$$= -\vec{a}_y \frac{E_0^{-}}{\eta_1} \mathrm{e}^{\mathrm{j}(\omega t + k_1 z)}$$

By comparison

$$\vec{H}^{+} = \vec{a}_y H_0^{+} \mathrm{e}^{\mathrm{j}(\omega t - k_1 z)}$$

$$= \vec{a}_y \frac{E_0^{+}}{\eta_1} \mathrm{e}^{\mathrm{j}(\omega t - k_1 z)}$$

Transmitted Wave



$$\vec{E}^{T} = \vec{a}_x E_x^{T} = \vec{a}_x E_0^{T} \mathrm{e}^{\mathrm{j}(\omega t - k_2 z)}$$

By comparison

$$\vec{E}^{+} = \vec{a}_x E_x^{+} = \vec{a}_x E_0^{+} \mathrm{e}^{\mathrm{j}(\omega t - k_1 z)}$$

By comparison

$$\vec{H}^{T} = \vec{a}_y H_0^{T} \mathrm{e}^{\mathrm{j}(\omega t - k_2 z)} = \vec{a}_y \frac{E_0^{T}}{\eta_2} \mathrm{e}^{\mathrm{j}(\omega t - k_2 z)}$$

$$\vec{H}^{+} = \vec{a}_y H_0^{+} \mathrm{e}^{\mathrm{j}(\omega t - k_1 z)} = \vec{a}_y \frac{E_0^{+}}{\eta_1} \mathrm{e}^{\mathrm{j}(\omega t - k_1 z)}$$

Reflection Coefficient of E-Field

$$R = \left.\frac{E_0^-}{E_0^+}\right|_{z=0}$$

$$\vec{E}^+ = \vec{a}_x E_0^+ \mathrm{e}^{\mathrm{j}(\omega t - k_1 z)}$$

$$\vec{E}^- = \vec{a}_x E_0^- \mathrm{e}^{\mathrm{j}(\omega t + k_1 z)}$$

$$\vec{E}^T = \vec{a}_x E_0^T \mathrm{e}^{\mathrm{j}(\omega t - k_2 z)}$$

$$\vec{H}^+ = \vec{a}_y \frac{E_0^+}{\eta_1} \mathrm{e}^{\mathrm{j}(\omega t - k_1 z)}$$

$$\vec{H}^- = -\vec{a}_y \frac{E_0^-}{\eta_1} \mathrm{e}^{\mathrm{j}(\omega t + k_1 z)}$$

$$\vec{H}^T = \vec{a}_y \frac{E_0^T}{\eta_2} \mathrm{e}^{\mathrm{j}(\omega t - k_2 z)}$$

Due to E_{1t}=E_{2t} $\quad E_{1t} = E_0^+ + E_0^- = E_{2t}$

Due to H_{1t}=H_{2t} $\quad H_{1t} = H_0^+ + H_0^- = \frac{E_0^+}{\eta_1} - \frac{E_0^-}{\eta_1} = H_{2t} = \frac{E_0^T}{\eta_2} = \frac{E_{2t}}{\eta_2}$

so

$$E_{2t} = E_0^+ + E_0^- = \frac{\eta_2}{\eta_1}(E_0^+ - E_0^-)$$

$$R = \frac{E_0^-}{E_0^+} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

Transmission Coefficient of E-Field

$$T = \left.\frac{E_0^T}{E_0^+}\right|_{z=0}$$

$$T = \frac{E_0^T}{E_0^+} = \frac{E_{2t}}{E_0^+} = \frac{E_{1t}}{E_0^+} = \frac{E_0^+ + E_0^-}{E_0^+} = 1 + R$$

$$T = \frac{2\eta_2}{\eta_2 + \eta_1}$$

$$T = \frac{2\sqrt{\varepsilon_1}}{\sqrt{\varepsilon_1} + \sqrt{\varepsilon_2}}$$

$\mu \approx \mu_0$

$$\eta = \sqrt{\frac{\mu}{\varepsilon}}$$

$$R = \left.\frac{E_0^-}{E_0^+}\right|_{z=0}$$

$$R = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

$$R = \frac{\sqrt{\varepsilon_1} - \sqrt{\varepsilon_2}}{\sqrt{\varepsilon_1} + \sqrt{\varepsilon_2}}$$

8.14 A uniform plane wave with electric field intensity $\boldsymbol{E}^{+} = E_0(\boldsymbol{e}_x + \mathrm{j}\boldsymbol{e}_y)\,\mathrm{e}^{-\mathrm{j}\beta z}$, is normally incident from air to a lossless dielectric board (it is dielectric for $z \geqslant 0$, its $\mu_r = 1$, $\varepsilon_r = 4$). Find:

(1) the electric field intensity of reflected wave;

(2) the magnetic field intensity of transmission wave;

(3) the respective polarization situation of incident wave, reflected wave and transmission wave.

3) 入射波、反射波和折射波电场各自的极化情况。

【分析】先求出 η_1，η_2，再用垂直入射时 R，T 的表示式求出 R，T 的值。然后，由 $\boldsymbol{E}^{-}=R\boldsymbol{E}^{+}$，$\boldsymbol{E}^{T}=T\boldsymbol{E}^{+}$，即可求出反射波及折射波电磁场，极化问题也就迎刃而解了。

【解】求 η_2，η_1，R，T：

$$\eta_2 = \sqrt{\frac{\mu}{\varepsilon}} = \frac{1}{\sqrt{4}}\eta_0 = \frac{1}{2}\eta_0 = 60\pi\ \Omega$$

$$\eta_1 = \eta_0 = 120\pi\ \Omega$$

反射系数为

$$R = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = -\frac{1}{3}$$

折射系数为

$$T = \frac{2\eta_2}{\eta_2 + \eta_1} = \frac{2}{3}$$

1) 反射波的电场强度为

$$\boldsymbol{E}^{-} = R\boldsymbol{E}^{+} = -\frac{1}{3}E_0\left(\boldsymbol{e}_x + \mathrm{j}\boldsymbol{e}_y\right)\mathrm{e}^{\mathrm{j}kz}$$

2) 折射波的电场强度为

$$\boldsymbol{E}^{T} = T\boldsymbol{E}^{+} = \frac{2}{3}E_0\left(\boldsymbol{e}_x + \mathrm{j}\boldsymbol{e}_y\right)\mathrm{e}^{-\mathrm{j}2kz}$$

折射波的磁场强度为

$$\boldsymbol{H}^{T} = \boldsymbol{e}_k \times \boldsymbol{E}^{T}/\eta_2 = \frac{1}{90\pi}E_0\left(\boldsymbol{e}_y - \mathrm{j}\boldsymbol{e}_x\right)\mathrm{e}^{-\mathrm{j}2kz}$$

式中，$\boldsymbol{e}_k$ 为波矢量 $\boldsymbol{k}$ 的单位矢量，即传播方向的 $\boldsymbol{e}_z$。

3) 入射波电场为左旋圆极化波，反射波电场为右旋圆极化波，折射波电场为左旋圆极化波。

【点评】折射波是在媒质 2 之中，它的相移常数 k_2 不等于入射波的相移常数 k，而是 $k_2=\omega\sqrt{\mu_0\varepsilon_2}=\omega\sqrt{\mu_0\varepsilon_0}\sqrt{4}=2k_0$。

Field and

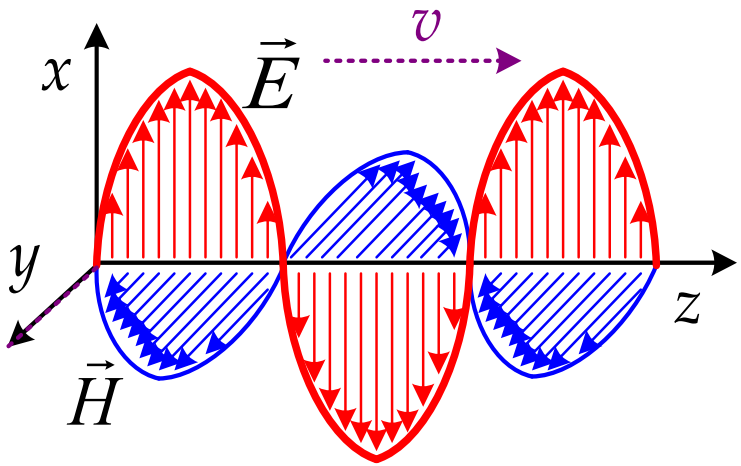
9.3 Oblique incidence on surface of perfect conductor

Wave along z-axis

$$\vec{E} = \vec{a}_x E_m^+ e^{-jkz}$$

$$\vec{H} = \vec{a}_y \frac{1}{\eta} \cdot E_m^+ \cdot e^{-jkz}$$

$$\vec{H} = \frac{1}{\eta} \vec{a}_z \times \vec{E}$$



Wave along x-axis

$$\vec{E} = \vec{a}_z E_m^+ e^{-jkx}$$

$$\vec{H} = \vec{a}_y \frac{1}{\eta} \cdot E_m^+ \cdot e^{-jkx}$$

$$\vec{H} = \frac{1}{\eta} \vec{a}_x \times \vec{E}$$

Wave along arbitrary direction

$$\vec{E} = \vec{a}_E E_m^+ e^{-j\vec{k} \bullet \vec{r}}$$

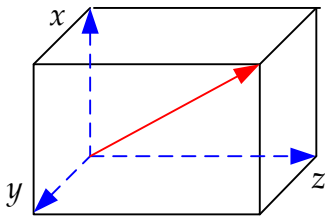
$$\vec{H} = \vec{a}_H \frac{1}{\eta} \cdot E_m^+ \cdot e^{-j\vec{k} \bullet \vec{r}}$$

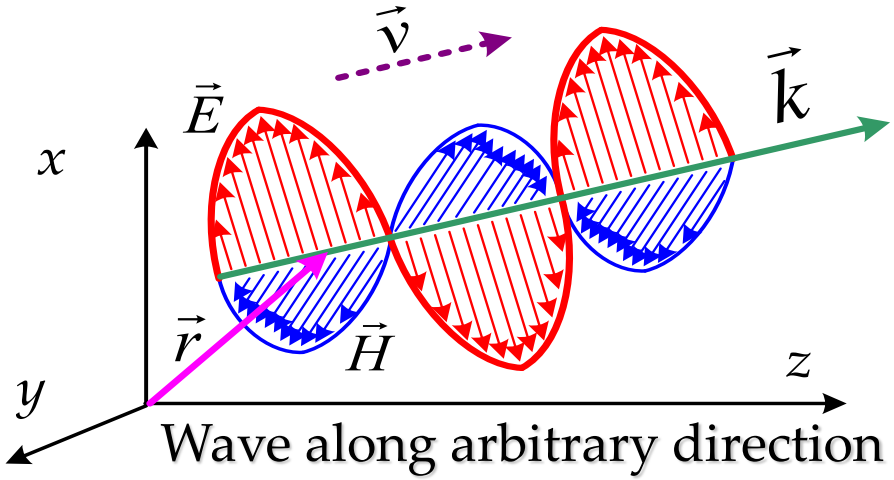
$$\vec{H} = \frac{1}{\eta} \vec{a}_k \times \vec{E} \qquad \vec{k} = \vec{a}_k k$$

Wave Vector $\vec{k}$

$$\vec{E} = \vec{a}_E E_m^+ e^{-j\vec{k}\bullet\vec{r}}$$

$$\vec{H} = \vec{a}_H \frac{1}{\eta} \cdot E_m^+ \cdot e^{-j\vec{k}\bullet\vec{r}}$$





$$\vec{\boldsymbol{k}} = \vec{\boldsymbol{e}}_x k_x + \vec{\boldsymbol{e}}_y k_y + \vec{\boldsymbol{e}}_z k_z = \vec{\boldsymbol{e}}_x k\cos\alpha + \vec{\boldsymbol{e}}_y k\cos\beta + \vec{\boldsymbol{e}}_z k\cos\gamma$$

$$k^2 = k_x^2 + k_y^2 + k_z^2$$

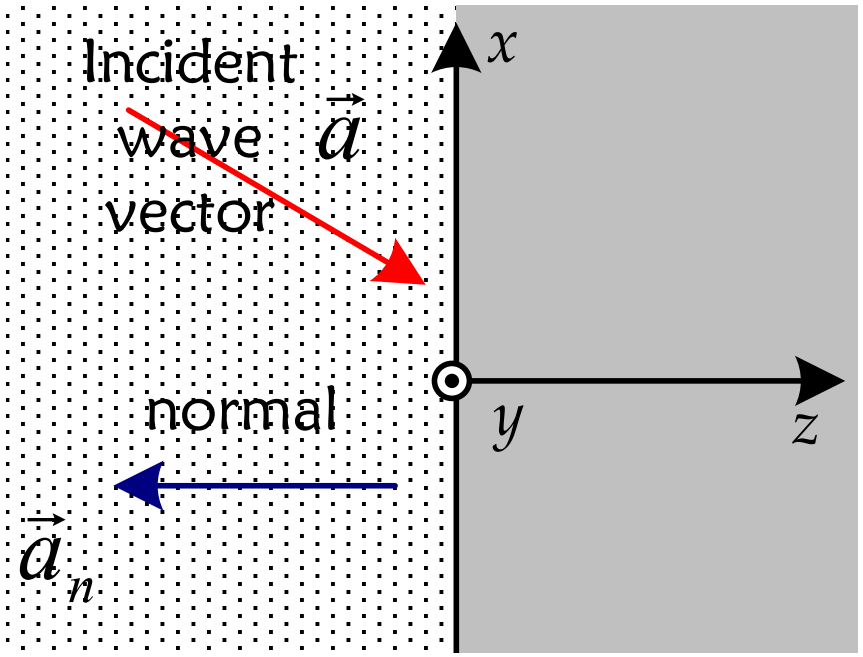
$$\vec{\boldsymbol{k}} \cdot \vec{r} = (\vec{\boldsymbol{e}}_x k_x + \vec{\boldsymbol{e}}_y k_y + \vec{\boldsymbol{e}}_z k_z) \cdot (\vec{\boldsymbol{e}}_x x + \vec{\boldsymbol{e}}_y y + \vec{\boldsymbol{e}}_z z)$$

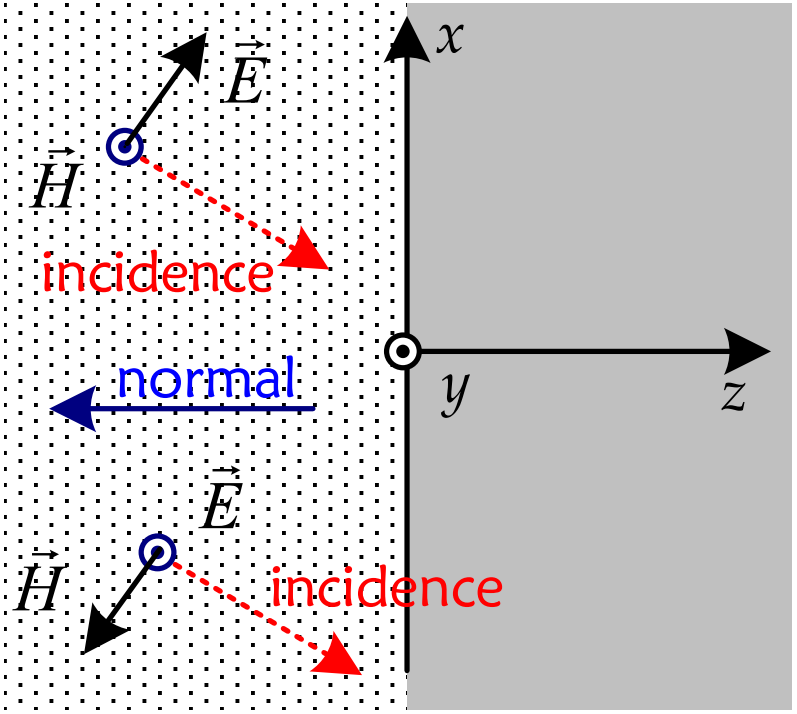
$$= k_x x + k_y y + k_z z$$

Classification of Oblique Incidence

Incident Plane is determined by wave vector & normal vector.

- parallel polarized incidence: E lies in incident plane
- normal polarized incidence: E is normal to incident plane





What kind of incidence?

normal polarized incidence

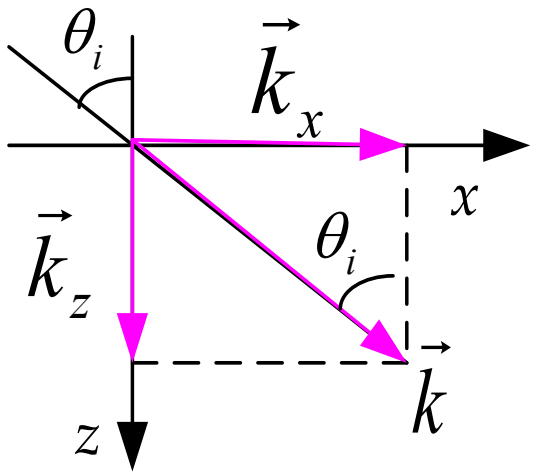
Incident Wave

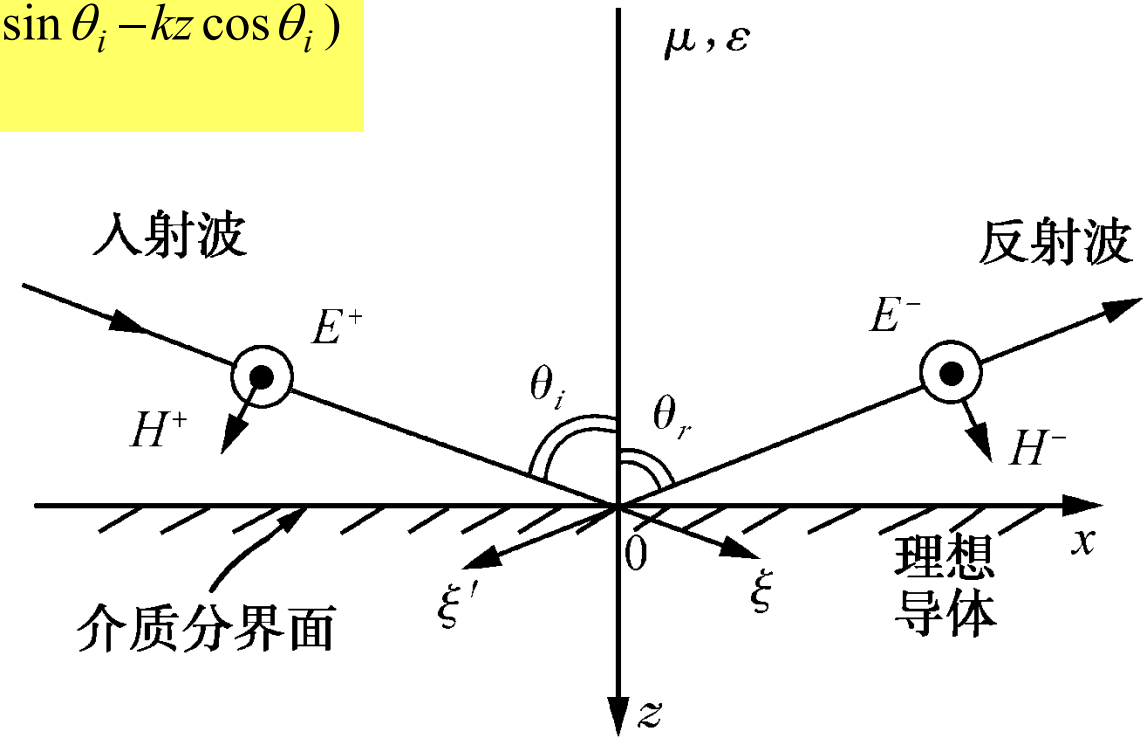
$$\vec{E}^{+} = \vec{e}_y E_y^{+} = \vec{e}_y E_0^{+} \mathrm{e}^{\mathrm{j}(\omega t - \vec{k}\bullet\vec{r})}$$

$$= \vec{e}_y E_0^{+} \exp\{\mathrm{j}(\omega t - k_x x - k_z z)\}$$

$$= \vec{e}_y E_0^{+} \mathrm{e}^{\mathrm{j}(\omega t - kx\sin\theta_i - kz\cos\theta_i)}$$

$$\vec{k}\cdot\vec{r} = k_x x + k_y y + k_z z$$



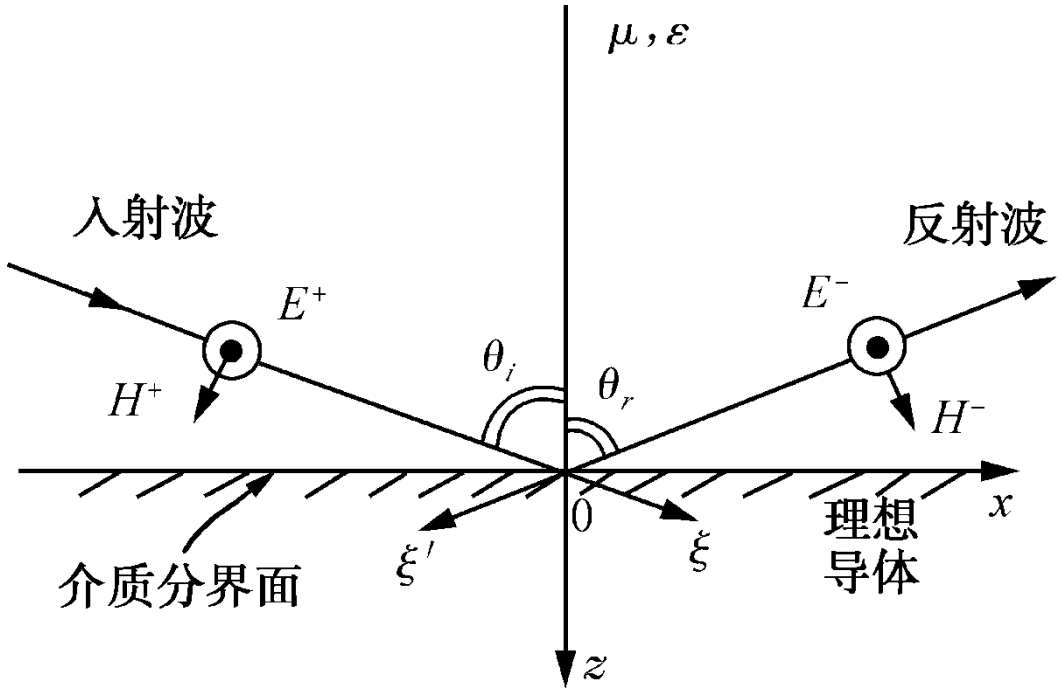


Reflected Wave

$$\vec{E}^- = \vec{e}_y E_y^-$$

$$= \vec{e}_y E_0^- \mathrm{e}^{\mathrm{j}(\omega t + \vec{k}^- \bullet \vec{r})}$$

$$= \cdots \quad ?$$

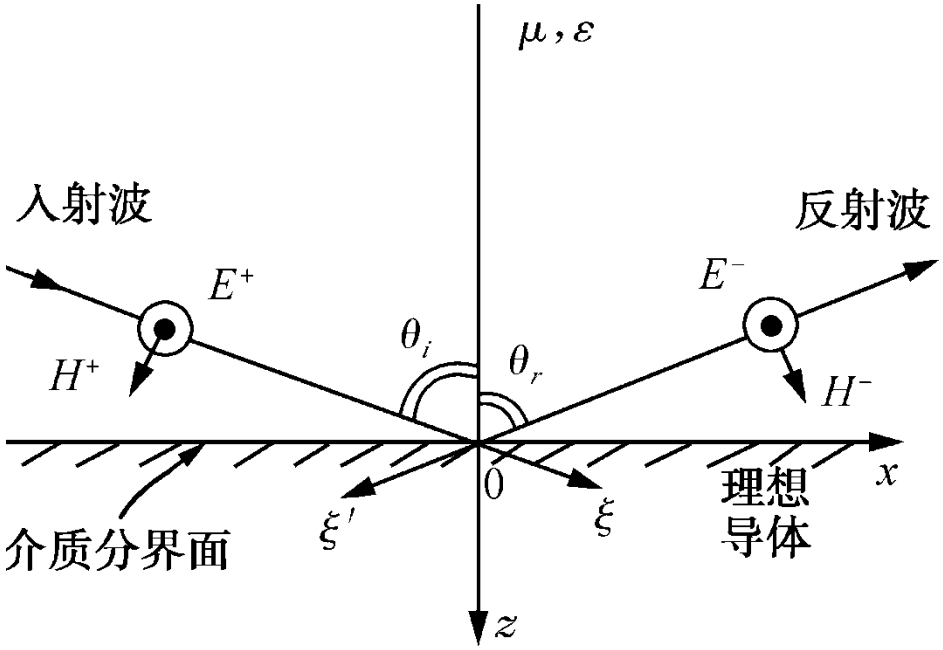


$$\vec{E}^- = \vec{e}_y E_0^- \mathrm{e}^{\mathrm{j}(\omega t - kx\sin\theta_r + kz\cos\theta_r)}$$

Total E-Field

$$\vec{E}^{+} = \vec{e}_y E_0^{+} \mathrm{e}^{\mathrm{j}(\omega t - kx\sin\theta_i - kz\cos\theta_i)}$$

$$\vec{E}^{-} = \vec{e}_y E_0^{-} \mathrm{e}^{\mathrm{j}(\omega t - kx\sin\theta_r + kz\cos\theta_r)}$$



$$E_y = E_y^{+} + E_y^{-}$$

$$= E_0^{+} \mathrm{e}^{\mathrm{j}(\omega t - k\sin\theta_i x - k\cos\theta_i z)} + E_0^{-} \mathrm{e}^{\mathrm{j}(\omega t - k\sin\theta_r x + k\cos\theta_r z)}$$

$$E_0^{+} \mathrm{e}^{\mathrm{j}k\sin\theta_i x} + E_0^{-} \mathrm{e}^{\mathrm{j}k\sin\theta_r x} = 0$$

E-field is tangential and thus continuous at boundary z=0.

Law of reflection $\theta_i = \theta_r = \theta$ $\qquad E_0^{+} = -E_0^{-}$

M-Field

$$\vec{\boldsymbol{H}}^{+} = \vec{\boldsymbol{e}}_x H_x^{+} + \vec{\boldsymbol{e}}_z H_z^{+}$$

where $H_x^{+} = -H_0^{+} \mathrm{e}^{\mathrm{j}(\omega t - k\sin\theta_i x - k\cos\theta_i z)} \cdot \cos\theta_i$

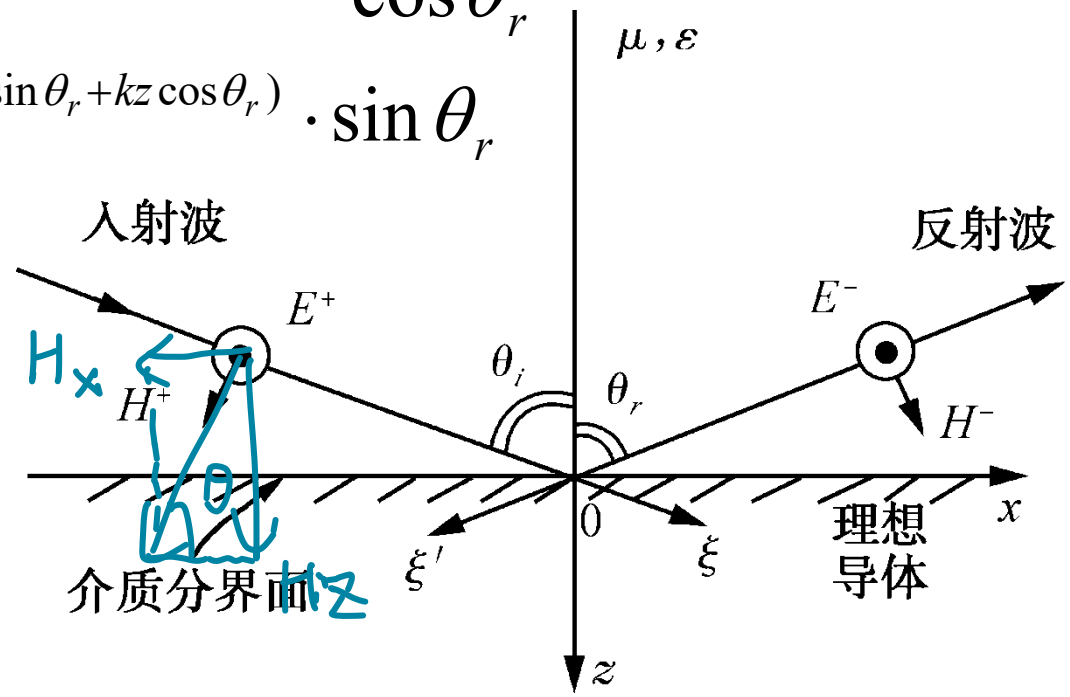
$$H_z^{+} = H_0^{+} \mathrm{e}^{\mathrm{j}(\omega t - k\sin\theta_i x - k\cos\theta_i z)} \cdot \sin\theta_i$$

$\theta_i = \theta_r = \theta$

$H_0^{+} = -H_0^{-}$

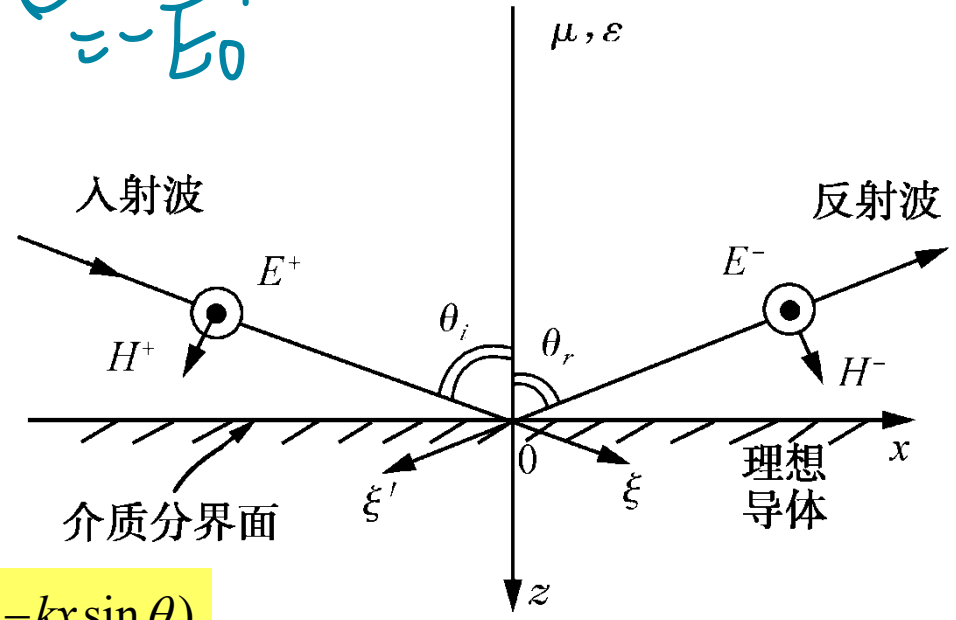
Similarly $H_x^{-} = H_0^{-} \mathrm{e}^{\mathrm{j}(\omega t - kx\sin\theta_r + kz\cos\theta_r)} \cdot \cos\theta_r$

$$H_z^{-} = H_0^{-} \mathrm{e}^{\mathrm{j}(\omega t - kx\sin\theta_r + kz\cos\theta_r)} \cdot \sin\theta_r$$



$$E_y = E_0^+ e^{j(\omega t - k\sin\theta_i x - k\cos\theta_i z)} + E_0^- e^{j(\omega t - k\sin\theta_r x + k\cos\theta_r z)}$$

$E_0^- = -E_0^+$



Standing wave along z-axis

Traveling wave along x-axis

$$E_y = -j2E_0^+ \sin(kz\cos\theta) e^{j(\omega t - kx\sin\theta)}$$

$$H_z = -j\frac{2E_0^+}{\eta}\sin\theta\sin(kz\cos\theta) e^{j(\omega t - kx\sin\theta)}$$

$$H_x = \frac{-2E_0^+}{\eta}\cos\theta\cos(kz\cos\theta) e^{j(\omega t - kx\sin\theta)}$$

垂直入射→橫电波

Only E lies in transverse plane, hence the name TE-wave.

Parallel Polarized Oblique Incidence

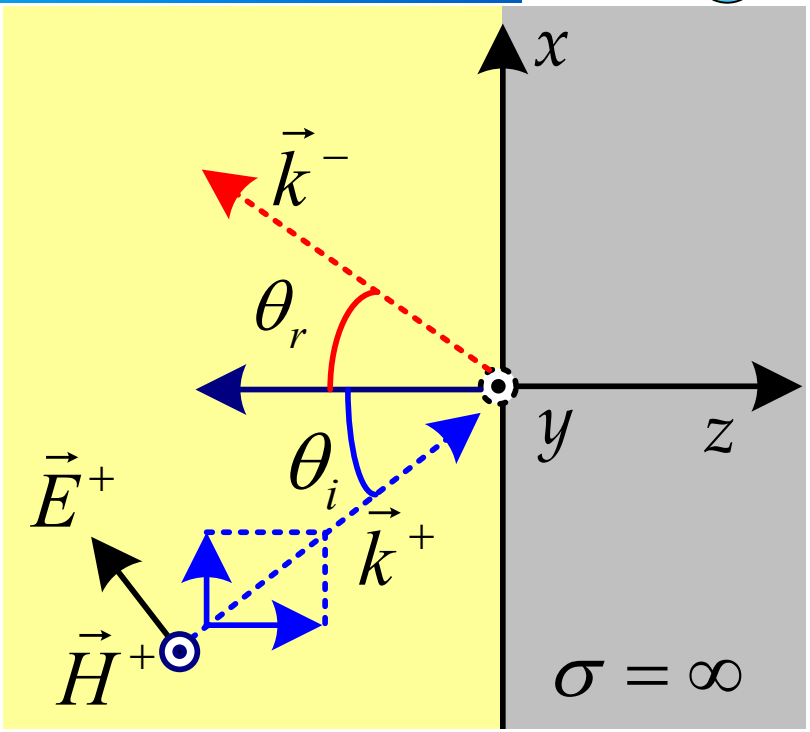
$$\vec{E}^{+} = \vec{E}_0^{+} \cdot e^{-j\vec{k}^{+} \bullet \vec{r}}$$

$$\vec{E}^{-} = \vec{E}_0^{-} \cdot e^{-j\vec{k}^{-} \bullet \vec{r}}$$

$$\vec{r} = \{x, y, z\}$$

$$\vec{k}^{+} = \vec{a}^{+} k = \vec{a}_x \left(k \cdot \sin\theta_i\right) + \vec{a}_z \left(k \cdot \cos\theta_i\right)$$

$$\vec{k}^{-} = \vec{a}^{-} k = \vec{a}_x \left(k \cdot \sin\theta_r\right) - \vec{a}_z \left(k \cdot \cos\theta_r\right)$$



At boundary $\vec{E}_1 \big|_{z=0} = 0$

Total fields: $\vec{E}_1 = \vec{E}^{+} + \vec{E}^{-}$

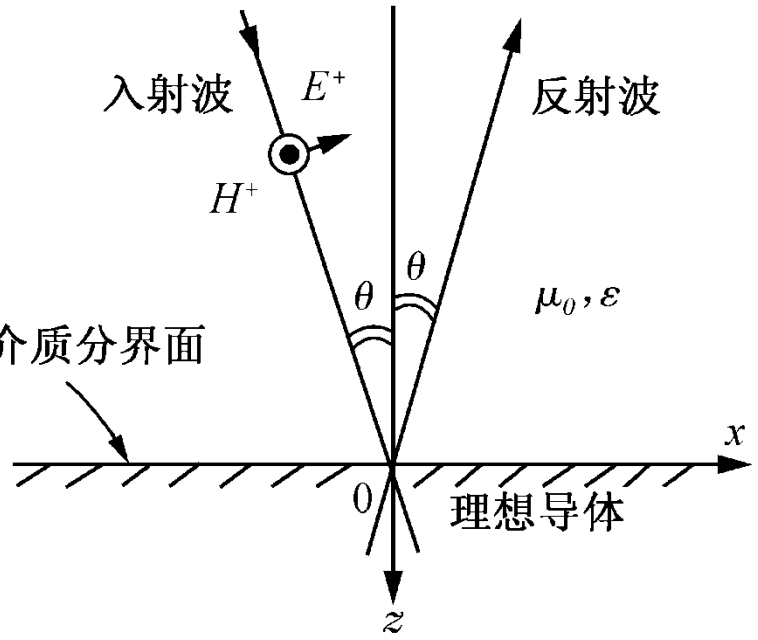
$$\vec{H}_1 = \vec{H}^{+} + \vec{H}^{-} = \frac{1}{\eta_1}\vec{a}^{+} \times \vec{E}^{+} + \frac{1}{\eta_1}\vec{a}^{-} \times \vec{E}^{-}$$

Total Field

$$E_z = 2E_0^+ \sin\theta \cos(kz\cos\theta)\mathrm{e}^{\mathrm{j}(\omega t - kx\sin\theta)}$$
$$= -2H_0^+ \eta \sin\theta \cos(kz\cos\theta)\mathrm{e}^{\mathrm{j}(\omega t - kx\sin\theta)}$$

$$E_x = -\mathrm{j}2E_0^+ \cos\theta \sin(kz\cos\theta)\mathrm{e}^{\mathrm{j}(\omega t - kx\sin\theta)}$$
$$= -\mathrm{j}2H_0^+ \eta \cos\theta \sin(kz\cos\theta)\mathrm{e}^{\mathrm{j}(\omega t - kx\sin\theta)}$$

$$H_y = 2H_0^+ \cos(kz\cos\theta)\mathrm{e}^{\mathrm{j}(\omega t - kx\sin\theta)}$$



Standing wave along z-axis

Traveling wave along x-axis

Only M-field lies in transverse plane, hence the name TM-wave.

Example 8.1 A TEM wave obliquely incident from the air upon an interface of media at $z = 0$, the electric field strength of incident wave is given as

$$E_y^+ = 5\mathrm{e}^{\mathrm{j}(-3x-2z)}$$

Try to find: (1) the angle of incidence θ_i;

(2) the unit vector e_{k_i} of the ray of incident wave.

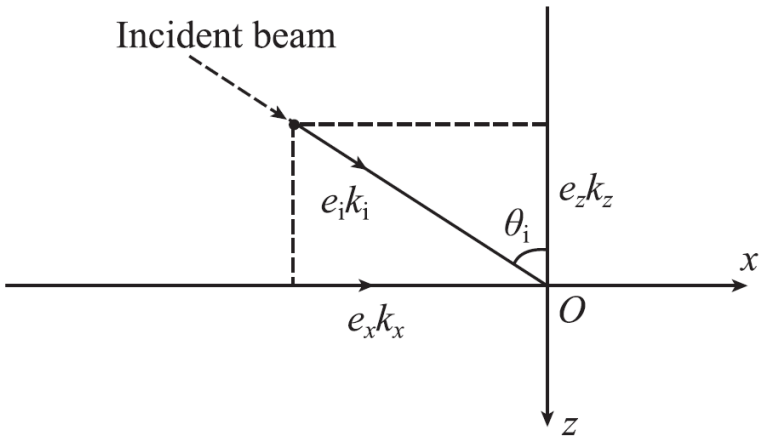


Figure of Example 8.1

Solutions (1) the angle of incidence θ_i.

From the figure, we have

$$\begin{cases} k_x = k_\mathrm{i}\sin\theta_\mathrm{i} = 3 \\ k_z = k_\mathrm{i}\cos\theta_\mathrm{i} = 2 \end{cases}$$

Thus

$$\tan\theta_\mathrm{i} = \frac{3}{2}$$

$$\theta_\mathrm{i} = 56.3^\circ$$

(2) The unit vector e_{k_i} of the ray of incident wave

Because

$$\boldsymbol{k}_\mathrm{i} = \boldsymbol{e}_x k_x + \boldsymbol{e}_y k_y + \boldsymbol{e}_z k_z = 3\boldsymbol{e}_x + 2\boldsymbol{e}_z$$

then

$$\boldsymbol{e}_{k_\mathrm{i}} = \frac{\boldsymbol{k}_\mathrm{i}}{|\boldsymbol{k}_\mathrm{i}|} = \frac{\boldsymbol{k}_\mathrm{i}}{\sqrt{3^2+2^2}} = \frac{3}{\sqrt{13}}\boldsymbol{e}_x + \frac{2}{\sqrt{13}}\boldsymbol{e}_z$$

8.5 A harmonically varying uniform plane wave is incident from air upon the surface of perfect conductor at $z=0$ (xOy plane), given that the incident electric field is: $\boldsymbol{E}^{+}=\boldsymbol{e}_y 10\mathrm{e}^{\mathrm{j}(\omega t-6x-8z)}\mathrm{V/m}$. Find:

(1) the incident angle θ_{i} of the wave;

(2) the frequency f and the wavelength λ;

(3) write the complex form of electric field for reflected wave;

(4) write the expression of the electric field of the total wave.

【解】由给定的入射波电场可知

$$k_x = k\sin\theta_i = 6 \tag{8.14.1}$$

$$k_z = k\cos\theta_i = 8 \tag{8.14.2}$$

1) 求入射角 θ_i。

由式(8.14.1)、式(8.14.2)得

$$\tan\theta_i = 3/4$$

$$\theta_i = 36.87^\circ$$

2) 求频率 f 及波长 λ。

因为

$$k_x = 6 = \frac{2\pi}{\lambda}\sin 36.87^\circ$$

所以

$$f = c/\lambda = 3\times 10^8/0.6283 = 477.465\times 10^6\ \mathrm{Hz}$$

3) 反射波电场的复数表示式。

因为入射波电场为 y 分量，与理想导体表面相平行，则电场的反射系数 $R=-1$，且反射波电场沿 $+x$，$-z$ 方向传播，所以有

$$\boldsymbol{E}^{-} = -\boldsymbol{e}_y 10\cdot \mathrm{e}^{\mathrm{j}(\omega t-6x+8z)}$$

4) 合成波电场。

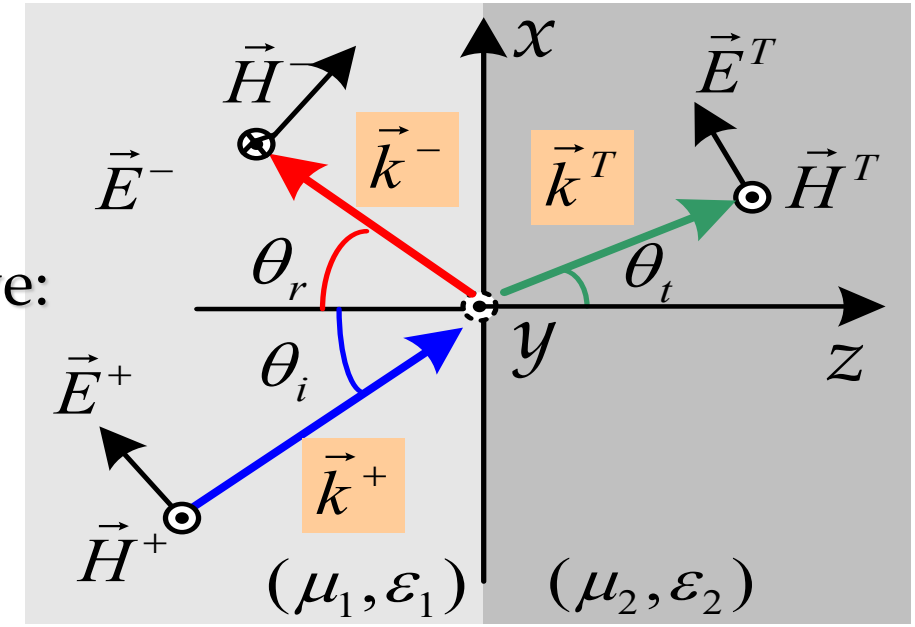
$$\boldsymbol{E} = \boldsymbol{E}^{+}+\boldsymbol{E}^{-} = \boldsymbol{e}_y\left\{10\cdot \mathrm{e}^{\mathrm{j}(\omega t-6x-8z)} - 10\cdot \mathrm{e}^{\mathrm{j}(\omega t-6x+8z)}\right\}$$

$$= \boldsymbol{e}_y 10\cdot \mathrm{e}^{\mathrm{j}(\omega t-6x)}\cdot\left\{\mathrm{e}^{-\mathrm{j}8z} - 10\cdot \mathrm{e}^{\mathrm{j}8z}\right\}$$

$$= -\mathrm{j}\boldsymbol{e}_y 20\sin(8z)\cdot \mathrm{e}^{\mathrm{j}(\omega t-6x)}$$

9.4.1 Snell's Law

- Assume the boundary is an infinite plane surface. Due to continuation of k_x on the boundary surface, we must have:



$$n_1 k_0 \sin\theta_i = n_1 k_0 \sin\theta_r = n_2 k_0 \sin\theta_t$$

$$k_0^2 = \omega^2 \mu_0 \varepsilon_0 \qquad n = \sqrt{\mu_r \varepsilon_r}$$

$$\therefore \theta_i = \theta_r \qquad \sin\theta_i = \frac{n_2}{n_1} \sin\theta_t$$

$$k_x^+ = k_x^- = k_x^T$$

- It is Snell's Law. Furthermore, the incident ray, reflected ray and transmitted ray must lies on the same plane.

Snell's Law

$$\theta_i = \theta_r \qquad \sin\theta_i = \frac{n_2}{n_1}\sin\theta_t$$

- Furthermore, the incident ray, reflected ray and transmitted ray must lies on the same plane.

$$n = \sqrt{\mu_r \varepsilon_r}$$ Index of the Material (折射率)

Assuming $\mu_1 \approx \mu_2 \approx \mu_0$

$$\sin\theta_i = \frac{\sqrt{\varepsilon_{r2}}}{\sqrt{\varepsilon_{r1}}}\sin\theta_t$$

Then we have

$$\theta_i = \theta_r \qquad \sin\theta_i = \frac{\sqrt{\varepsilon_2}}{\sqrt{\varepsilon_1}}\sin\theta_t$$

9.4.3. total reflection & total transmission

重点

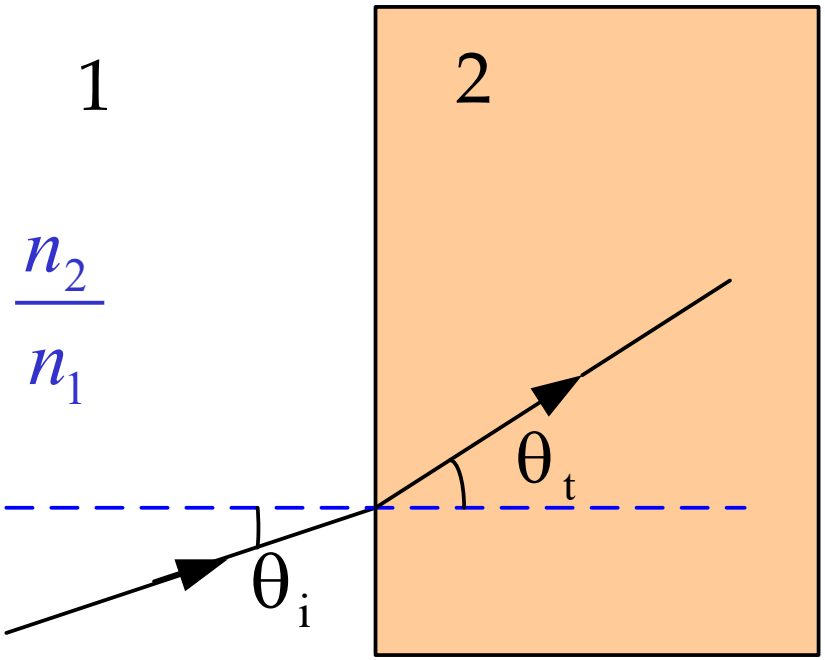
Condition of total reflection (from ray optics point of view):

$\theta_t = 90^o$

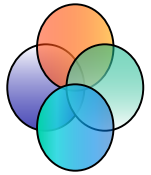
$$\sin\theta_i = \frac{\sqrt{\varepsilon_2}}{\sqrt{\varepsilon_1}}\sin 90^o = \frac{\sqrt{\varepsilon_2}}{\sqrt{\varepsilon_1}} = \frac{n_2}{n_1}$$

$$\sin\theta_c = \frac{n_2}{n_1}$$

$$\theta_c = \sin^{-1}\frac{n_2}{n_1} = \text{Critical Angle}$$



A Summary: Condition of total reflection

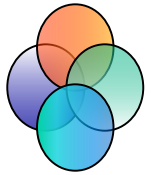


- Incident angle θ_i lies in interval of $[\theta_c, 90^o]$.
- If incident wave goes from region 1 to region 2,

$$\varepsilon_2 < \varepsilon_1 \quad \text{判斷題} \quad \sin\theta_c = \frac{\sqrt{\varepsilon_2}}{\sqrt{\varepsilon_1}} = \frac{n_2}{n_1}$$

$$n_2 < n_1$$

9.4.4. Total Transmission & Brewster Angle



Condition: R=0 or $|T|$=1

For normal polarized incidence

$$R_{(N)} = \frac{\cos\theta_i - \sqrt{\left(\frac{\varepsilon_2}{\varepsilon_1}\right) - \sin^2\theta_i}}{\cos\theta_i + \sqrt{\left(\frac{\varepsilon_2}{\varepsilon_1}\right) - \sin^2\theta_i}} = 0$$

$$T_{(N)} = \frac{2\cos\theta_i}{\cos\theta_i + \sqrt{\left(\frac{\varepsilon_2}{\varepsilon_1}\right) - \sin^2\theta_i}} = \pm 1$$

是非判断选择题

These 2 equations have no solution.

Therefore, normal-polarized plane wave is unable to transmit totally from dielectric ① to dielectric ②.

垂直极化平面波无论怎么入射都不可能形成全折射！

Condition: $R=0$ or $|T|=1$

For parallel polarized incidence

$$R_P = \frac{-\left(\frac{\varepsilon_2}{\varepsilon_1}\right)\cos\theta_i + \sqrt{\left(\frac{\varepsilon_2}{\varepsilon_1}\right) - \sin^2\theta_i}}{\left(\frac{\varepsilon_2}{\varepsilon_1}\right)\cos\theta_i + \sqrt{\left(\frac{\varepsilon_2}{\varepsilon_1}\right) - \sin^2\theta_i}} = 0$$

$$T_P = \frac{2\sqrt{\left(\frac{\varepsilon_2}{\varepsilon_1}\right) - \sin^2\theta_i}}{\left(\frac{\varepsilon_2}{\varepsilon_1}\right)\cos\theta_i + \sqrt{\left(\frac{\varepsilon_2}{\varepsilon_1}\right) - \sin^2\theta_i}} = \pm 1$$

Solving these 2 equations, we have

$$\tan\theta_i = \sqrt{\varepsilon_2}\,/\sqrt{\varepsilon_1}$$

$$\text{or} \quad \tan\theta_i = n_2/n_1$$

$$\therefore \theta_i = \theta_B = tg^{-1}\, n_2/n_1$$

θ_B is named as Brewster Angle.

Example 有关斜入射的应用

- f=300MHz的均匀平面波从$\mu_1=\mu_0$, $\varepsilon_1=4\varepsilon_0$的媒质中入射到与空气的分界面上
- 求: (1) 若想全反射, 入射角应该是多大?
  (2) 若入射波是圆极化波, 且只希望反射波是单一的线极化波, 入射角应该是多大?

解: (1) 回忆: 形成全反射的2个条件

- 入射角 θ_i 落在区间 $[\theta_c, 90^o]$ 内
- 若电磁波从 1区 入射 2区 形成全反射, 则必须 $\varepsilon_2 < \varepsilon_1$ $n_2 < n_1$

$$\text{由题意}: \theta_c = \arcsin\frac{n_2}{n_1} = \arcsin\frac{\sqrt{\varepsilon_2}}{\sqrt{\varepsilon_1}} = \arcsin\frac{1}{2} = 30^o$$

答: 入射角 θ_i 落在区间 $[30^o, 90^o]$ 内

- f=300MHz的均匀平面波从$\mu_1=\mu_0$, $\varepsilon_1=4\varepsilon_0$的媒质中入射到与空气的分界面上
- 求: (2) 若入射波是圆极化波, 且只希望反射波是单一的线极化波,入射角应该是多大?

解: (2) 圆极化波可以分解成2个极化方向相垂直的线极化波.

那么, 在什么情况下可将两种线极化波分开?

垂直极化平面波不可能形成全折射!

平行极化平面波可以发生全折射:

$$\theta_i = \theta_B = \arctan\frac{\sqrt{\varepsilon_2}}{\sqrt{\varepsilon_1}}$$

答: 入射角 θ_i=26.57°

8.16 求光线自玻璃($n=1.5$)到空气的临界角和布儒斯特角。并证明在一般情况下,临界角 θ_C 总大于布儒斯特角 θ_B。

【解】因为 $\sin\theta_C=\frac{n_2}{n_1}=\frac{1}{1.5}$,所以

$$\theta_C = 41.81^\circ$$

因为

$$\tan\theta_B = \frac{n_2}{n_1} = \frac{1}{1.5}$$

所以

$$\theta_B = 33.69^\circ$$

一般情况下,由于

$$\sin\theta_B = \sqrt{\frac{\varepsilon_2}{\varepsilon_1+\varepsilon_2}}, \sin\theta_C = \sqrt{\frac{\varepsilon_2}{\varepsilon_1}}$$

所以

$$\theta_C > \theta_B$$

$$\frac{E_x}{H_y}=\sqrt{\frac{\mu}{\varepsilon}}=\eta$$

A uniform plane wave is normally incident from the air to the surface (z=0) of a perfect dielectrics ($\varepsilon_r=64\quad \mu_r=1$). If the reflected electric field is $\dot{\vec{E}}^{-}=\vec{e}_x E_0 e^{j\beta z}$, determine: [9 marks]

1) the reflection *R* and transmission coefficient T.

2) the expressions of incidence electric field $\dot{\vec{E}}^{+}$ and transmitted electric field $\dot{\vec{E}}^{T}$.

$$R=\frac{\eta_2-\eta_1}{\eta_2+\eta_1}=\frac{\frac{1}{8}-1}{\frac{1}{8}+1}=-\frac{7}{9}$$

$$T=1+R=\frac{2}{9}$$

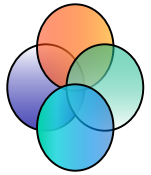
$$\vec{E}^{-}=\vec{e}_x E_0 e^{j(\omega t+\beta z)}$$

$$\vec{E}^{+}=-\vec{e}_x\frac{9}{7}E_0 e^{j(\omega t-\beta z)}$$

$$\beta'=\sqrt{\varepsilon_r}\beta=8\beta$$

$$\vec{E}^{T}=-\vec{e}_x\frac{2}{7}E_0 e^{j(\omega t-8\beta z)}$$

3 Types of Electromagnetic Waves



波导

10分计算

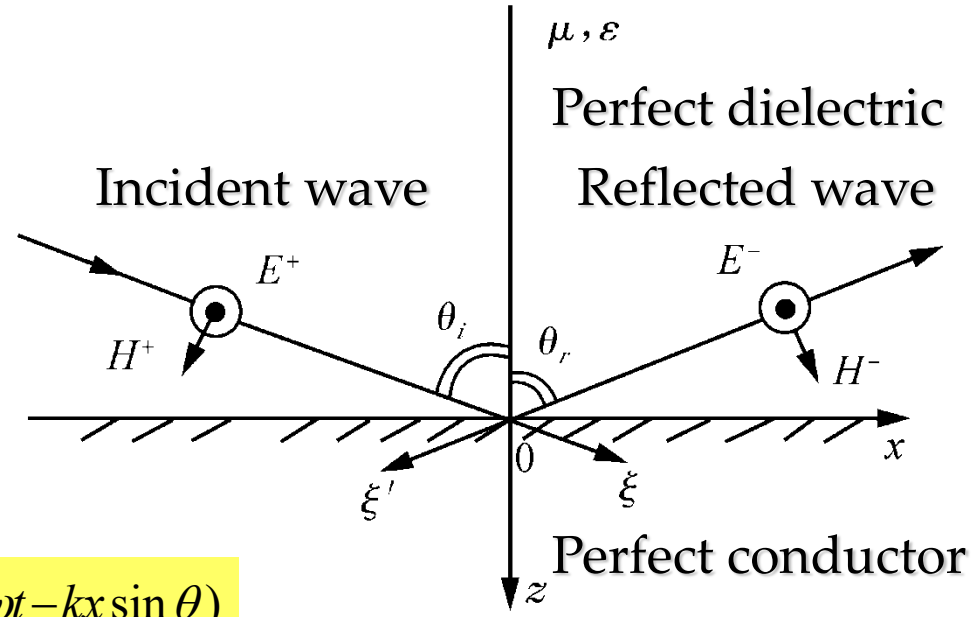
6-8分

① TEM Wave: Transverse Electromagnetic Wave
② TM Wave: Transverse Magnetic Wave,
also known as E-Wave
③ TE Wave: Transverse Electric Wave,
also known as H-Wave

Transverse means there is no field component along the direction of wave propagation.

Oblique incidence of a perpendicularly polarized wave

Along the x direction, TE waves



$$E_y = -\mathrm{j}2E_0^+ \sin(kz\cos\theta)\mathrm{e}^{\mathrm{j}(\omega t - kx\sin\theta)}$$

$$H_z = -\mathrm{j}\frac{2E_0^+}{\eta}\sin\theta\sin(kz\cos\theta)\mathrm{e}^{\mathrm{j}(\omega t - kx\sin\theta)}$$

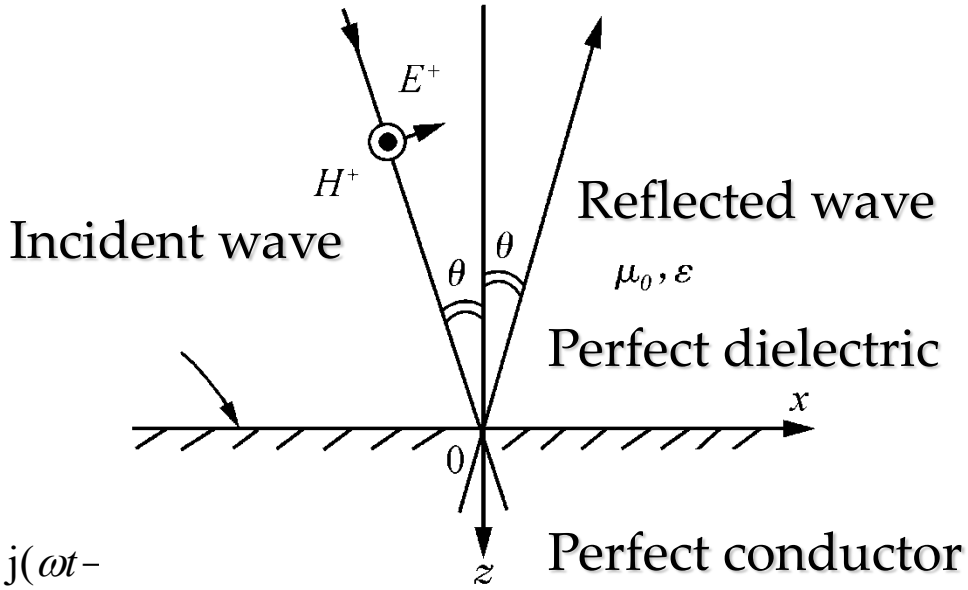
Along the z direction, standing wave

$$H_x = \frac{-2E_0^+}{\eta}\cos\theta\cos(kz\cos\theta)\mathrm{e}^{\mathrm{j}(\omega t - kx\sin\theta)}$$

Along the x direction, traveling wave

Oblique incidence of a parallel polarized wave

Along the x direction, TM waves



$$E_z = -2H_0^+ \eta \sin\theta \cos(kz\cos\theta) e^{j(\omega t -}$$

$$E_x = -j2H_0^+ \eta \cos\theta \sin(kz\cos\theta) e^{j(\omega t - kx\sin\theta)}$$

Along the z direction, standing wave

$$H_y = 2H_0^+ \cos(kz\cos\theta) e^{j(\omega t - kx\sin\theta)}$$

Along the x direction, traveling wave

$$E_z = B_o \sin\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)e^{j(\omega t-\beta z)}$$

$$E_x = -j\frac{\beta}{k_c^2}\frac{m\pi}{a}B_o \cos\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)e^{j(\omega t-\beta z)}$$

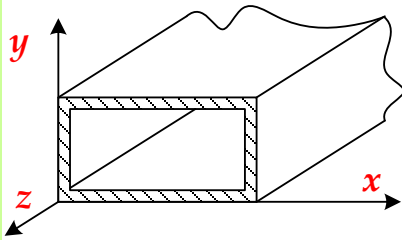
$$E_y = -j\frac{\beta}{k_c^2}\frac{n\pi}{b}B_o \sin\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)e^{j(\omega t-\beta z)}$$

$$H_x = j\frac{\omega\varepsilon}{k_c^2}\frac{n\pi}{b}B_o \sin\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)e^{j(\omega t-\beta z)}$$

$$H_y = -j\frac{\omega\varepsilon}{k_c^2}\frac{m\pi}{a}B_o \cos\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)e^{j(\omega t-\beta z)}$$

$$H_z = 0$$

$$k_c^2 = k_x^2 + k_y^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$



TM wave

$$H_z = A_o \cos\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)e^{j(\omega t-\beta z)}$$

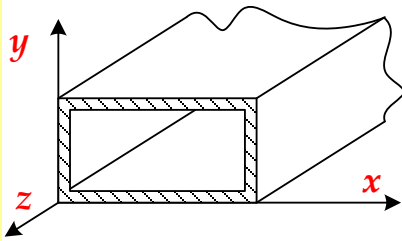
$$H_x = j\frac{\beta}{k_c^2}\frac{m\pi}{a}A_o \sin\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)e^{j(\omega t-\beta z)}$$

$$H_y = j\frac{\beta}{k_c^2}\frac{n\pi}{b}A_o \cos\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)e^{j(\omega t-\beta z)}$$

$$E_x = j\frac{\omega\mu}{k_c^2}\frac{n\pi}{b}A_o \cos\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{b}y\right)e^{j(\omega t-\beta z)}$$

$$E_y = -j\frac{\omega\mu}{k_c^2}\frac{m\pi}{a}A_o \sin\left(\frac{m\pi}{a}x\right)\cos\left(\frac{n\pi}{b}y\right)e^{j(\omega t-\beta z)}$$

$$E_z = 0$$



TE wave

$$k_c^2 = k_x^2 + k_y^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$

$$k^2 = k_c^2 - \gamma^2$$

$$\gamma^2 = k_c^2 - k^2 \geq 0$$

The wave is said to be evanescent.

$$\therefore k_c^2 \geq k^2 = (2\pi/\lambda)^2$$

$$\therefore k_c^2 = k_x^2 + k_y^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \geq \left(\frac{2\pi}{\lambda}\right)^2$$

The cutoff wavelength

$$\therefore \lambda \geq \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

$$\lambda_c = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

$$f_c = \frac{v}{\lambda_c} = \frac{1}{\lambda_c}\frac{c}{\sqrt{\varepsilon_r \mu_r}}$$

$\lambda < \lambda_c$ or $f > f_c$, the wave can propagate.

$$k^2 = k_c^2 - \gamma^2$$

$$\gamma^2 = k_c^2 - k^2 \geq 0$$

The wave is said to be evanescent.

$$\therefore k_c^2 \geq k^2 = (2\pi/\lambda)^2$$

$$\therefore k_c^2 = k_x^2 + k_y^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \geq \left(\frac{2\pi}{\lambda}\right)^2$$

$$\therefore \lambda \geq \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

The cutoff wavelength

$$\lambda_c = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

$$f_c = \frac{v}{\lambda_c} = \frac{1}{\lambda_c}\frac{c}{\sqrt{\varepsilon_r \mu_r}}$$

$\lambda < \lambda_c$ or $f > f_c$, the wave can propagate.

The phase constant β

If lossless, then $\gamma = j\beta$ $\quad \gamma^2 = (j\beta)^2 = -\beta^2 = k_c^2 - k^2$

$$\beta^2 = k^2 - k_c^2 \xrightarrow{2\pi = k\lambda = k_c\lambda_c} \beta^2 = k^2 - k^2\frac{\lambda^2}{\lambda_c^2}$$

$$\beta = k\sqrt{1-(\lambda/\lambda_c)^2}$$

The Waveguide wavelength: $\lambda_g = \dfrac{2\pi}{\beta} = \dfrac{2\pi}{k\sqrt{1-(\lambda/\lambda_c)^2}} = \dfrac{\lambda}{\sqrt{1-(\lambda/\lambda_c)^2}}$

The phase velocity

$$v_p = \frac{\omega}{\beta} = \frac{\omega}{k\sqrt{1-(\lambda/\lambda_c)^2}} = \frac{v}{\sqrt{1-(\lambda/\lambda_c)^2}}$$

The group velocity v_g

$$v_g = \frac{d\omega}{d\beta} = \frac{1}{d\beta/d\omega}$$

$$2\pi = k\lambda = k_c\lambda_c$$

$$\beta = \sqrt{k^2 - k_c^2} = \sqrt{\left(\frac{\omega}{v}\right)^2 - \left(\frac{2\pi}{\lambda_c}\right)^2}$$

$$v_g = v\sqrt{1-(\lambda/\lambda_c)^2}$$

Thus, a waveguide acts as a dispersive medium.

3. The lowest-order mode & single-mode propagation

The lowest-order mode is the one with the maximum cutoff wavelength.

$$\lambda_c = \frac{2\pi}{k_c} = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

When $\lambda < \lambda_c$, the wave can propagate.

The cutoff wavelengths of TE_{mn} modes

When $a = 2b$,

$$\lambda_{c(10)} = 2a \quad \text{for } TE_{10}$$

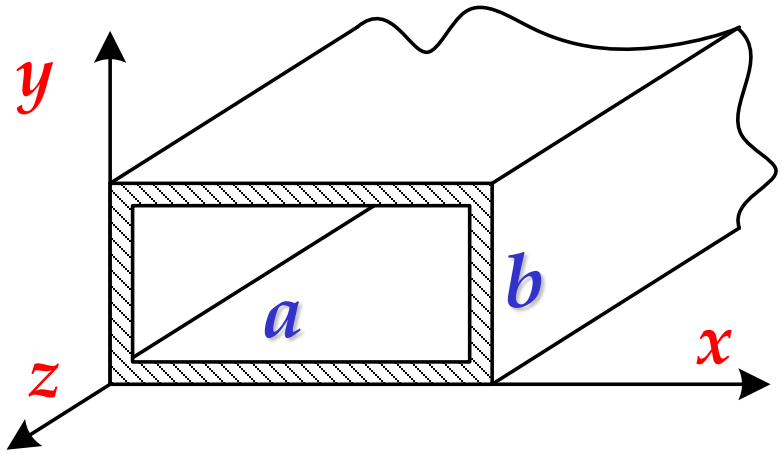
$$\lambda_{c(20)} = a \quad \text{for } TE_{20}$$

$$\lambda_{c(01)} = 2b = a \quad \text{for } TE_{01}$$

$$\lambda_{c(11)} = 2a/\sqrt{5} \quad \text{for } TE_{11}$$

$$\lambda_{c(30)} = 2a/3 \quad \text{for } TE_{30}$$

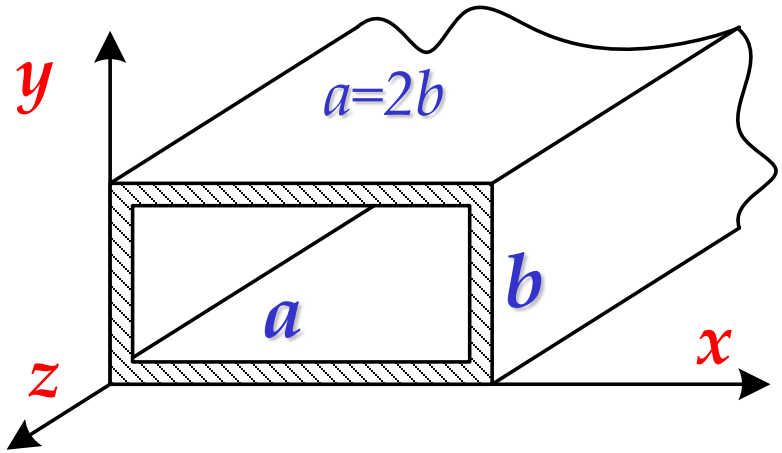
$$\lambda_{c(02)} = b = a/2 \quad \text{for } TE_{02}$$



$$\lambda_c = \frac{2\pi}{k_c} = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

The maximum cutoff wavelength of TE waves

$$\lambda_{c(10)} = 2a$$



TE_{10} mode is the lowest-order mode in all TE modes.

Likewise, for TM waves

必考

$$\lambda_{c(11)} = 2a/\sqrt{5}$$

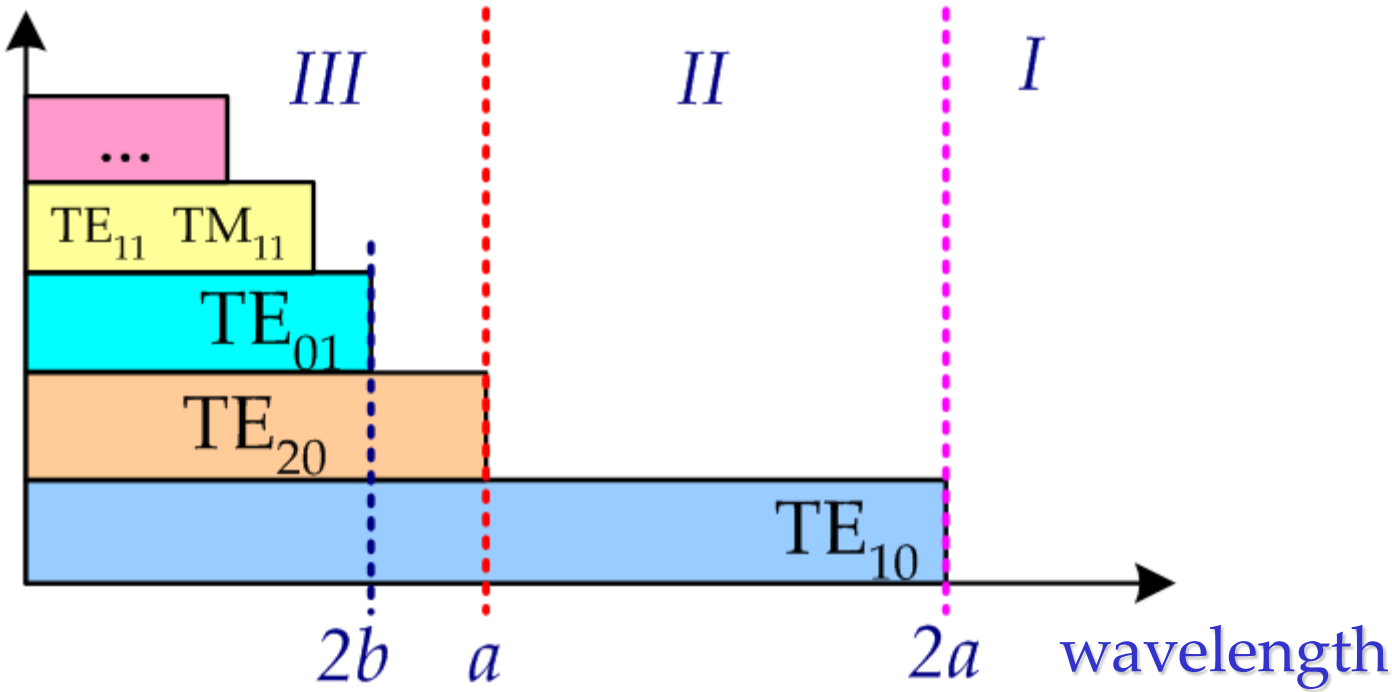
TM_{11} mode is the lowest-order mode in all TM modes.

Therefore,

When $a=2b$, the lowest-order mode is TE_{10}.

$$\lambda_{c(10)} = 2a$$

The distribution of cutoff wavelength



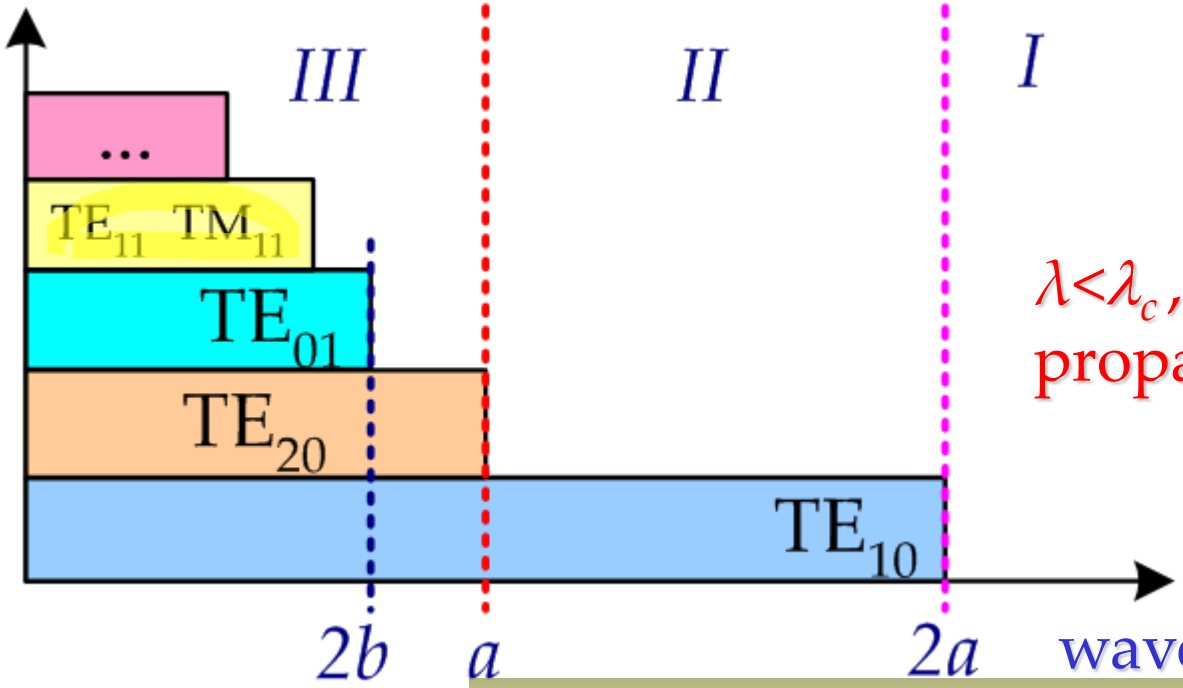
When $\lambda<\lambda_c$, the wave can propagate.

Region I：cutoff——evanescent

Region II：single-mode ——only one mode can exist

Region III：multi-mode——more than one mode can exist

The conditions of single-mode propagation

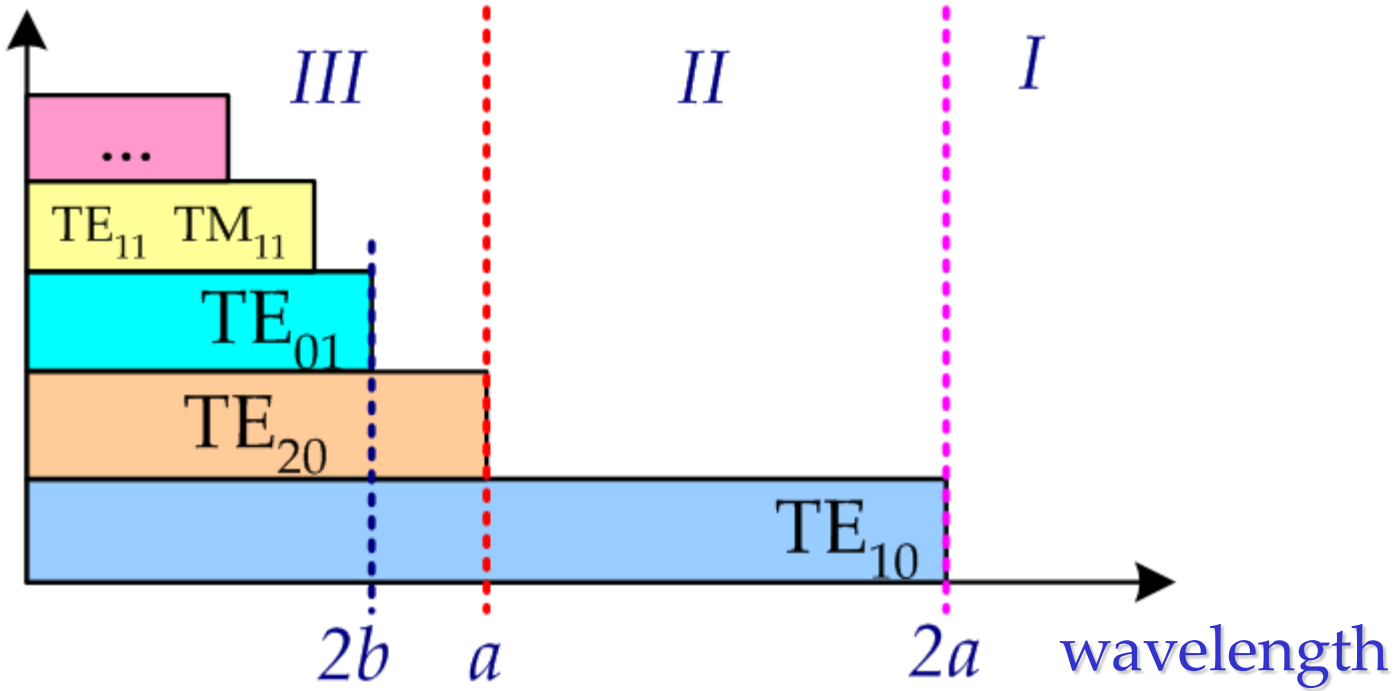


$\lambda<\lambda_c$, the wave can propagate.

$$\begin{cases} 2a > \lambda > a \\ \lambda/2 < a < \lambda \quad (a \geq 2b) \end{cases}$$

TE_{10} is the only propagating mode, called the dominant mode.

Mode degeneracy



Mode degeneracy: two modes have the same cutoff wavelength.

For example, TE_{11} and TM_{11} are the degenerate mode,

but have different distribution.

Parameters of TE_{10} Mode

The cutoff frequency:

$$f_c = \frac{v}{\lambda_c} = \frac{1}{2a}\frac{c}{\sqrt{\varepsilon_r \mu_r}}$$

The cutoff wavelength:

$$\lambda_c = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}} = 2a$$

The phase constant:

$$\beta = k\sqrt{1-(\lambda/\lambda_c)^2} = k\sqrt{1-(\lambda/2a)^2}$$

The waveguide wavelength:

$$\lambda_g = \frac{2\pi}{\beta} = \frac{\lambda}{\sqrt{1-(\lambda/2a)^2}}$$

The phase velocity:

$$v_p = \frac{\omega}{\beta} == \frac{v}{\sqrt{1-(\lambda/2a)^2}}$$

The group velocity:

$$v_g = v\sqrt{1-(\lambda/\lambda_c)^2} = v\sqrt{1-(\lambda/2a)^2}$$

The wave impedance:

$$Z_{W(TE)} = \frac{\eta}{\sqrt{1-(\lambda/\lambda_c)^2}} = \frac{\eta}{\sqrt{1-(\lambda/2a)^2}}$$

10.2 已知矩形波导的截面尺寸为 $a\times b=23\ \text{mm}\times 10\ \text{mm}$，试求当工作波长 $\lambda=10\ \text{mm}$ 时，波导中能传输哪些模式？$\lambda=30\ \text{mm}$ 呢？

【分析】该题目考查的是对矩形波导“截止波长”的计算以及“简并”模式这两个知识点。

【解】波导中能传输得波形必须满足“高通”的条件，即

$$f>(f_c)_{mn}\ \text{或}\ \lambda<(\lambda_c)_{mn} \tag{10.2.1}$$

对于矩形波导，截止波长的表示式为

$$\lambda_c=\frac{2\pi}{\sqrt{\left(\frac{m\pi}{a}\right)^2+\left(\frac{n\pi}{b}\right)^2}} \tag{10.2.2}$$

依据传输条件

$$\lambda<(f_c)_{mn}=\frac{2}{\sqrt{\left(\frac{m}{a}\right)^2+\left(\frac{n}{b}\right)^2}} \tag{10.2.3}$$

1) 已知：$\lambda=10\ \text{mm}$，$a=23\ \text{mm}$，$b=10\ \text{mm}$，代入式(10.2.3)，化简得到

$$\left(\frac{m}{23}\right)^2+\left(\frac{n}{10}\right)^2<0.04 \tag{10.2.4}$$

能满足传输条件的 m 和 n 为

① $m=0$ 时，$n<2$，所以能传导如下波型：TE_{01}。

② $m=1$ 时，$n<1.95$，所以能传导如下波型：TE_{11}，TM_{11}，TE_{10}。

③ $m=2$ 时，$n<1.8$，所以能传导如下波型：TE_{21}，TM_{21}，TE_{20}。

④ $m=3$ 时，$n<1.5$，所以能传导如下波型：TE_{30}，TE_{31}，TM_{31}。

⑤ $m=4$ 时，$n<0.95$，所以能传导如下波型：TE_{40}。

⑥ $m=5$ 时，n 无解，所以能传导该模式的电磁波。

共能传输 11 种波型。

多模

枚举法

2) 当 $\lambda=30\ \text{mm}$ 时，式(10.2.3)化简得

$$\left(\frac{m}{23}\right)^2+\left(\frac{n}{10}\right)^2<\frac{4}{900} \tag{10.2.5}$$

单模

① $m=0$ 时，$n<0.67$，所以不能传导相应波长的模式(无波型存在)。

② $m=1$ 时，$n<0.5$，所以能传导如下波型：有 TE_{10}。

③ $m=2$ 时，n 无解，所以不能传导相应波长的模式(无波型存在)。

共能传输 1 种波型。

10.5 Rectangular waveguide with $a = 7\text{cm}$, $b = 3\text{cm}$, which is filled with air or lossless medium with $\varepsilon_r = 4$, $\mu_r = 1$.

(1) Find the cutoff frequency and guided wavelength for TE_{10}, TE_{20}, TE_{01}, TE_{11}, TM_{11} modes.

(2) Which modes can propagate at $f = 3 \times 10^9 \text{Hz}$.

$$f_c = \frac{v}{\lambda_c} = \frac{1}{\lambda_c} \frac{c}{\sqrt{\varepsilon_r \mu_r}}$$

$$\lambda_c = \frac{2}{\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}}$$

10.5 (1) When filled with air, λ_c are 14, 7, 6, 5.51, 5.51cm respectively
f_c are 2.14, 4.29, 5.00, 5.44×10^9Hz respectively

When filled with medium, λ_c are 14, 7, 6, 5.51, 5.51cm respectively
f_c are 1.07, 2.14, 2.50, 2.72×10^9Hz respectively

(2) When filled with air, there is only TE_{10} mode; When filled with medium, there are TE_{10}, TE_{20}, TE_{01}, TE_{11}, TM_{11} totally 5 modes.

用图解法是一种比较直观的方法。

10.7 已知在 2cm×2cm 的方形中空波导管中传播频率为 10GHz 的 TE_{10} 波。

1）试求传播矢量 $\boldsymbol{k}$。

2）写出电磁场各分量的表达式。

3）求波导波长。

4）求电磁波的相速和群速。

【分析】首先，波矢量的概念要清楚：$\boldsymbol{k}=\boldsymbol{e}_x k_x+\boldsymbol{e}_y k_y+\boldsymbol{e}_z k_z$；真空中 10GHz 的微波信号，波长 $\lambda_0=3\text{cm}$，传输的是基模 TE_{10}。

【解】1）方形波导，$a=b=0.02\text{m}$，波矢 $\boldsymbol{k}$ 的各分量为

$$k_x=\frac{m\pi}{a}=50\pi \text{ rad/m}$$

$$k_y=\frac{n\pi}{b}=0 \text{ rad/m}$$

$$k_z=\sqrt{\omega^2\mu_0\cdot\varepsilon_0-k_x^2-k_y^2}=\sqrt{\frac{\omega^2}{c^2}-k_x^2-k_y^2}$$

$$=\sqrt{\left(\frac{2\pi}{\lambda_0}\right)^2-k_x^2-k_y^2}\approx 44\pi \text{ rad/m} \qquad (10.7.1)$$

2）TE_{10} 波场分量。

直接可以写出

$$E_x=E_z=0, E_y=E_0\sin(k_x x)\mathrm{e}^{\mathrm{j}(\omega t-k_z z)} \qquad (10.7.2)$$

通过麦克斯韦方程可以求出磁场的各个分量：

$$H_x=-\frac{k_z}{\omega\mu_0}E_0\sin(k_x x)\mathrm{e}^{\mathrm{j}(\omega t-k_z z)}, H_y=0, H_z=\frac{\mathrm{j}k_x}{\omega\mu_0}E_0\cos(k_x x)\mathrm{e}^{\mathrm{j}(\omega t-k_z z)}$$

3）波导波长。

$$\lambda_g=\frac{2\pi}{k_z}=0.045 \text{ m} \qquad (10.7.3)$$

该电磁波在自由空间中的波长为

$$\lambda_0=\frac{2\pi}{k_0}=\frac{c}{f}=0.03 \text{ m} \qquad (10.7.4)$$

不难看出 $\lambda_g>\lambda_0$。

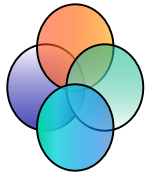
4）相速和群速。

$$v_p=\frac{\omega}{k_z}=f\cdot\lambda_g=4.5\times10^8 \text{ m/s} \qquad (10.7.5)$$

利用公式

$$v_p\cdot v_g=c^2 \qquad (10.7.6)$$

得



EXAMPLE 10.5

An air-filled waveguide operates at a frequency of 10 GHz. If the dimensions of the waveguide are $a = 2$ cm and $b = 1$ cm, determine the mode of propagation in the waveguide. 0102 0101

Solution Let us first consider the two lowest-order modes, TE_{01} and TE_{10}, and calculate the cutoff frequencies for both of them.

$$TE_{01}: \quad f_{c01} = \frac{u_p}{2b} = \frac{3 \times 10^8}{0.02} = 1.5 \times 10^{10} \text{ Hz} \quad \text{or} \quad 15 \text{ GHz}$$

Because the operating frequency ($f = 10$ GHz) is smaller than $f_{c01} = 15$ GHz, TE_{01} cannot be a possible propagating mode.

$$TE_{10}: \quad f_{c10} = \frac{u_p}{2a} = \frac{3 \times 10^8}{2 \times 2 \times 10^{-2}} = 0.75 \times 10^{10} \text{ Hz} \quad \text{or} \quad 7.55 \text{ GHz}$$

Therefore, TE_{10} is the only mode that can propagate in the waveguide.